

Anomaly Detection in Industrial Control Systems Using XAI and Contextual Severity Analysis

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Declaration

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This work has not been previously accepted in substance for any degree and is not being concurrently submitted in candidature for any degree.

Statement 2

This thesis is the result of my own investigations, except where otherwise stated. Other sources are acknowledged by citations giving explicit references. A bibliography is appended.

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Abstract

Industrial Control Systems (ICS) represent a type of critical infrastructures that control key processes like power grid and manufacturing operations. Due to how highly automated these systems are and how dependent on networked sensors and actuators they remain, these control systems remain especially vulnerable targets against cyberattacks. Although recent work has improved ICS-level anomaly detection and explainability, a central limitation continues to remain: no standardized approach to analysing severity of detected anomalies. Absent severity analysis, operators remain without clear recommendations about how promptly they should act, or which events carry the greatest level of safety threat.

This project extends the WaXAI (Explainable Anomaly Detection in Industrial Control Systems and Water Systems) framework by introducing a safety-focused severity scoring system. The system measures the criticality of the observed anomalies on three key dimensions: (i) safety criticality of the impacted component, (ii) stage significance, and (iii) operational disruption, capturing the degree of deviation of typical plant functionality. Our framework is trained and tested against ground-truth attack data available at the SWaT testbed, ensuring correspondence between observed events and actual safety consequences. Our principal deliverables are: (i) a systematic algorithm for scoring attack severities, (ii) attack ranking by severity for human operators, and (iii) a human-interpretable notification format enabling quick response and decision-making.

It is observed that this approach facilitates anomaly detection by taking another step beyond finding if an attack has occurred to determining how severe and disruptive it is. Using this, this research addresses one of the future work suggestions in WaXAI by developing a consistent manner to prioritize response, reduce operator workload, and improve ICS environment resilience.

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Chapter 1

1.1 Introduction

Industrial Control Systems (ICS) are the operational backbone of critical infrastructures such as waste and water systems, power systems, oil refineries, and transportation. Historically, the design philosophy of ICS has focused on reliability and availability rather than on security, especially cybersecurity. The effectiveness of Industry 4.0 and the emerging cyber-physical technologies have significantly increased both enhancements and vulnerabilities. ICS environments are interconnected, minimally human in the loop, and generate large volumes of real-time data enabled by sensors and actuators (Kayan et al., 2022).

The trend towards increased automation distorts the ICS systems, making them entice for adversaries. Since the well-documented attacks of Stuxnet (Iran, 2010), Triton/Trisis (Saudi Arabia, 2017), and more recently, the Colonial Pipeline (USA, 2021), it is clear that cyberattacks targeting ICS systems can not only cripple essential services, leading to potential financial and environmental harm, but in the worst-case scenarios may lead to the loss of human life. Furthermore, in instances of water treatment systems, adversarial influence targeting pumps, valves or chemical dosing will ultimately undermine public health and safety (Kundu et al., 2024).

In recent years, machine learning-based anomaly detection has become an important defensive cyber security mechanism. Anomaly detection models can deviate from a pre-filled rule, or signature, and instead learn what the normal operating characteristics of an ICS look like, so that when they learn that the model has deviated from these characteristics, they may be able to utilize anomaly detection and softly categorise these deviations as potential attacks (Nabi et al., 2025). WaXAI (Mathuros et al., 2024) is one model that has undertaken an approach that uses the secure water treatment dataset (SWaT), utilizing unsupervised models - ECOD, and DeepSVDD - to detect anomalies in conjunction with explainable artificial intelligence (XAI) techniques, e.g., SHAP and LIME, for model explainability and understandability. While WaXAI showed successful results in transparency and detection accuracy, we left an important gap uncovered. That is, anomaly detection frameworks do not report if any attack is severe, where operators need help to make prioritisation decisions.

This dissertation takes up the new research opportunity that WaXAI did not and proposes a severity scoring framework to quantify how critical or a system anomaly is, based upon the

importance of the safety of the component, that sustained for a day or period of anomalous behaviour (for example), and plant disruption. Therefore, this research turns anomaly detection from a binary attack/no attack decision, to a prioritised response, or safety-based response.

1.2 Motivation

Anomaly detection has proven to be an effective method of identifying attacks; however, it presents the challenge of managing onslaughts of alerts for operators of industrial control systems (ICS) (McCarthy et al., 2020). A water treatment plant may have numerous compromised notifications arrive at once and a few sensors dealing with abnormal readings. The lack of a systematic way of determining which alerts are the most relevant makes it possible for operators to significantly underreact to potential threats or overreact to non-serious deviations. This issue is worse in ICS contexts, in which safety and service continuity are paramount and a momentary lapse in response can mean cascading failures (Mohanty et al., 2024).

The WaXAI framework had already set the stage for explainability of anomaly detection results, making it easier to trust alerts from machine learning processes. The future work section of the paper acknowledges that explainability alone does not imply reliability. It is important that operators understand there is a layer of intelligence placed upon it. This represents whether there is an indication of severity about an attack which helps not only to understand whether an attack is present but also to determine the error is important, causing a degradation of human safety, impacting operations, and/or is active long enough to potentially become a more serious and relevant threat.

I am motivated to help close that gap with this dissertation. By measuring the severity of anomalies systematically, operators can:

- Prioritise harmful attacks above minor annoyances,
- Manage resources effectively during incident response,
- Diminish false positives without being unprepared,
- Strengthen confidence in the automated detection technology by providing some actionable decision support.

Therefore, severity scoring has the potential to turn anomaly detection into decision support, thus allowing ICS environments to move from detection to resilience.

1.3 Aims and Objectives

The overarching aim of this dissertation is to design, implement, and validate a safety-focused severity scoring framework for ICS cyberattacks, thereby extending the WaXAI approach with actionable intelligence for operators.

To achieve this aim, the dissertation pursues the following objectives:

1. Develop a severity scoring methodology that combines three dimensions:
 - Safety criticality of affected components,
 - Stage Importance of each component in the process, and
 - Operational disruption to plant processes.
2. Validate the scoring system by aligning detected anomalies with known ground-truth attacks.
3. Classify and rank attack scenarios by severity to demonstrate how prioritisation can be achieved.
4. Present results in a structured reporting and notification format suitable for practical operator use.

1.4 Summary

The structure of the dissertation mimics the following pattern:

Chapter 1: Introduction – The motivation, aims and objectives of the dissertation is discussed in this section.

Chapter 2: Background Research - It offers the initial ICS security background, the description of the anomaly detection methodologies, together with the research gap of the severity scoring.

Chapter 3: Literature Review - scans the work on anomaly detection, Explainable AI, and severity scoring paradigms.

Chapter 4: Methodology and Implementation - Details the dissertation workflow, ranging from dataset preprocessing up to deployments of the anomaly detecting models, XAI integration, and the construction of the severity scoring framework.

Chapter 5: Results - Displays the results of the framework of severity scoring, i.e., results of classification, stage-wise results, comparison between the performance of the proposed method and case studies.

Chapter 6: Difficulties, Limitations and Future Research - Details the most significant difficulties and the vulnerabilities of the research and suggests directions for the future to improve as well as generalize the framework.

Chapter 2

Background Research

Industrial Control Systems (ICS) are the operating heart of numerous critical infrastructures such as water treatment plants, power plants, oil refining plants, and factories. ICS integrate sensors and actuators, programmable logic controllers (PLCs), and supervisory mechanisms to make physical processes operate efficiently and safely, sometimes with little or no human intervention (Nankya et al., 2023). Historically, ICS were not connected to external networks and were engineered to have availability and reliability as ultimate requirements. Yet, due to the advent of Industry 4.0, ICS have digitally become more interconnected using corporate networks, cloud platforms, and remote monitoring platforms. Although increase in connectivity improves efficiency and performance, it also introduces ICS to a new set of cyber threats (Aoun et al., 2021).

This chapter aims to put the context around this dissertation. It discusses the composition and value of ICS in critical infrastructure, considers the cybersecurity challenges that make ICS attractive to adversaries, considers anomaly detection as an important defence strategy, reviews the datasets used to evaluate methods for detection, discusses the role of explainable AI (XAI) in creating operator confidence, and discusses the decision making issues for operators that create the motivation for developing a framework for severity scoring.

2.1 Industrial Control Systems and Critical Infrastructure

Industrial Control Systems have the function of measuring and actuating physical processes on a large scale (Dunn et al., n.d.). The sensors sense quantities such as water level, flow rates, pressures, and concentrations of chemicals, and actuators actuate the system by opening and closing valves, switching pumps on and off, or operating dosing mechanisms. Individual control is normally offered by virtue of PLCs that execute pre-decided logic, and Supervisory Control and Data Acquisition (SCADA) systems provide centralised control and allow human operators to act at times. On large or distributed sites, Distributed Control Systems (DCS) control collections of numerous site-wide controllers (Dalesio et al., 1994).

ICS lies at the heart of critical infrastructure. A water treatment station relies on ICS to supply safe, drinkable water continuously at a constant flow rate to households and industries. Power stations rely on ICS for distribution, balancing of loads, and stability of the system. Petroleum

and gas refining plants use control systems to automatize risky chemicals processes, and failure would result in gigantic and catastrophic explosions (Weiss, 2013). Similarly, transportation networks rely on ICS for signals and automation of traffic. Since these systems reside at the cyber and physical boundary, failures do not just affect information flows; they inflict environmental damage, economic loss, and indeed dangers to human life (Aziz et al., 2017).

2.2 Cybersecurity Challenges in ICS

Security challenges for ICS mostly stem from the shift from air-gapped to networked installations. Most plants previously utilized air-gapped networks, and they inherently accounted for defence against cyber-attack. The digital maturity trend, remote operations, and predictive maintenance have nonetheless networked ICS with corporate informational technology (IT) networks and, in most cases, the internet (Qin et al., 2019). The attack surface has thus exponentially increased.

ICS security contrasts from conventional IT security in these keyways:

- **Availability over Confidentiality:** confidentiality of information is preferred in most of the IT systems, but ICS prioritize the continuity of having processes kept continuously operational. As any temporary shutdown(downtime) causes disruption of critical services such as water supply or electric supply, it is prioritized (Green et al., 2016).
- **Legacy Components:** Some ICS use legacy components and software that have not been designed against current cyber security needs and, more significantly, possess an exploitation potential greater than the newer components (McLaughlin et al., 2016).
- **Restricted Patching Cycles:** Unlike IT systems that would easily have high-frequency patching, ICS is not patched often without disrupting activities. At times, patching is withheld for the sake of stability (Lindsay, 2013).
- **Stringent Real-Time Constraints:** ICS must respond in real-time to orders. Adding layers of conventional IT security adds the possibility of latency and creating hazardous operating conditions (McLaughlin et al., 2016).

The need for defenders to face these challenges is evident from real-world assaults. The 2010 Stuxnet worm, for instance, specially targeted Iran's nuclear plants' PLCs, dictating centrifuge behaviour while masking the attack from operators. The Triton (or Trisis) malware attack compromised safety instrumented systems at a Saudi Arabian petrochemical facility in 2017,

potentially causing a disastrous accident (Stergiopoulos et al., 2020). More recently, in 2021, the Oldsmar water treatment in Florida was infiltrated by attackers attempting to spike sodium hydroxide levels to lethal concentrations. Later in the year, the ransomware attack on the Colonial Pipeline suspended supplies of fuel across the eastern United States, demonstrating how highly reliant IT and OT systems now are. The examples demonstrate potential for devastation of ICS assaults and the importance of smart defence measures (Matos, 2021).

2.3 Anomaly Detection in ICS

Anomaly detection is important to the fight against ICS. While signature-based intrusion detection relies on pre-defined signatures of recognized attack modus operandi, anomaly detection observes patterns of deviation from normal practice. It is especially worth its value in ICS, in which attack methodologies would be new or zero-day, thus not taking any signature but inclined to compromise the dynamism of the system (Moorthy Agoramoorthy et al., 2023).

Anomalous detection methodologies tend to fall into the following:

- Statistical approaches: These estimate normal distribution and mark data points strongly deviating from it as anomalies (Rousseeuw & Hubert, 2017).
- Standard machine learning models: Algorithms like Support Vector Machines (SVM) and Isolation Forest try to find anomalies using learned decision boundaries (Agyemang, 2024).
- Deep learning models: Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Autoencoders learn temporal dependencies and nonlinearities of process data (Yu et al., 2020).
- Light, unlabelled networks like Empirical Cumulative Distribution Outlier Detection (ECOD) and Deep Support Vector Data Description (DeepSVDD) offer an excellent trade-off between the performance of the detection as well as the complexity of the computation (Essien & Giannetti, 2020).

A key facilitator for doing this work has been the availability of realistic datasets. Among the ones available, the Secure Water Treatment (SWaT) dataset is considered the de facto standard as it emulates a six-stage water treatment system with pumps, valves and chemical dosing equipment. The dataset consists of few days of normal operation and includes 41 cyber-attacks, each labelled with the component that was compromised and the duration of the attack (Lamshöft et al., 2021). Quality datasets like SWaT challenge others in the same vein, such as

the WADI (Water Distribution Testbed) dataset and BATADAL, for providing researchers with real-world situations to test and compare detection approaches.

2.4 Explainability and Operator Trust

Anomaly models have improved in precision but there is a nagging problem remaining: interpretability. Most of the top-performing models, especially those that leverage deep learning, act like black boxes: they provide an anomaly score but do not provide insight into why features contributed most strongly to the decision. In ICS, lack of explainability stifles operator confidence since safety decisions or expensive shutdowns cannot be made from opaque predictions (Alharbi et al., 2021).

Explainable AI (XAI) emerged exactly for such a challenge by bringing feature attributions. The most notable methodologies are:

- SHAP (Shapley Additive Explanations): Calculating feature contributions to anomalous scores using cooperative game theory (Li et al., 2024).
- LIME (Local Interpretable Model-Agnostic Explanations): Creates surrogate models for providing per-sample prediction explanations (Zafar & Khan, 2021).
- ALE (Accumulated Local Effects): Computes mean feature contributions and reduces correlation errors due to highly correlated features (Okoli, 2024).
- Integrated Gradients (IG): Assigns importance using gradients and is feasible for deep learning models (Qi et al., 2020).

WaXAI system showed that by combining anomaly detection and these XAI methods, localisation of the anomalies is enhanced and operators become more trusting of the system's results. Explainability by itself, however, does not give you a solution for prioritising. XAI gives you why and where things are anomalous but not how critical they are, and operators must use judgement and judge how severe they are (Asaye et al., 2025).

2.5 Decision-Making and the Need for Severity Scoring

In actual industrial plants, operators initially must contend with hundreds of alarms. The vast majority of these turn out to be false alarms, and the few remaining ones correspond to anomalies of limited operational significance. This situation, called alert fatigue, reduces the sensitivity of alarms and increases critical warning odds of being ignored. Even among valid

alarms, the criticality of anomalies is not necessarily the same and may span from minor deviations of redundant sensors to serious intrusions of safety-critical pumps. Treating all such alarms with the same urgency level is not practical or safe (Goel et al., 2017).

In information technology security, such challenges have been alleviated by assignments of severity ratings from frameworks like the Common Vulnerability Scoring System (CVSS) (Mell et al., 2007). These integrate exploitability and impact to give a score used to inform patching and incident response. CVSS, however, is not suitable for ICS due to its inability to reflect the physical and safety implications of process anomalies. Good ICS-specific framework, on the other side, needs to account for:

- Safety importance of the component: recognizing that pumps, valves, and dosing systems may be many times more critical for safety than sensors that are not critical.
- Stage importance of the system: accounting for the fact that errors on stages with more importance (i.e., chemical dosing) are more impactful than stages with less importance.
- Disruption of operations: quantification of how much an anomaly disrupts continuity, effectiveness or operational stability.

With these additional dimensions, a severity scoring system would allow operators to manage their response by focusing attention and resource on the most immediate of threats. This takes us from detection of anomalies in real time through decision support in a time frame that is practicable, making the ICS not simply secure but resilient to disruption.

Chapter 3

Literature Review

The preceding chapter introduced the value of Industrial Control Systems (ICS) in critical infrastructure, their growing susceptibility to cyberattacks, and operators' challenges with current dependence on anomaly detection and explainability. The current chapter shifts from descriptive to analytical: it critically investigates scholarly efforts in three focal areas of this dissertation. First, it surveys anomaly detection techniques used on ICS, ranging from historical statistical means to newer deep learning and lightweight unsupervised models. Secondly, it evaluates the use of explainable artificial intelligence (XAI) for rendering detection models understandable and interpretable to operators. Lastly, it looks at existing severity and risk evaluation frameworks, especially those from IT security, and examines their appropriateness and shortcomings in the realm of ICS. By bringing to attention the strengths and weaknesses of preexisting work, the current chapter sees the void that the current dissertation fills: the lack of a domain-specific severity score framework converting anomaly detection into decision support actionable.

3.1 Anomaly Detection in ICS

3.1.1 Traditional Statistical and Machine Learning Approaches

The early anomaly detection methods of ICS branched from statistical modelling and conventional machine learning. Principal Component Analysis (PCA) was first applied as one of the early methods by reducing high-dimensional sensor inputs into a low set of components and triggering alarms on deviations beyond normal variance. Though PCA is computationally efficient, its application of linear correlation limits its performance for nonlinear correlations common in industrial plants (Rassam et al., 2013).

Other popular classifiers include Support Vector Machines (SVM) and its extension, One-Class SVM (OC-SVM). By learning the boundary of normal data points, they detect outliers that lie beyond such a region (Zhu et al., 2016). The appeal of OC-SVM for ICS datasets with unlabelled anomalies was short-lived due to its quick achievement of scalability problems on the lengthy, high-rate time series of SWaT. k-Nearest Neighbours (kNN) methods, that classify by approximation of distance from points, were successful on small-scale test sets but computationally expensive for real-time ICS monitoring. Isolation Forests tried to overcome

these shortcomings by considering outliers as those points that are easily isolated with the fewest splits of a decision tree. While promising, they tended to incorrectly classify normal process variations as attacks, producing unacceptable false positives (Arslan et al., 2024).

These traditional methods made anomalous detection a possible defence but revealed glaring shortcomings. They were interpretable and lightweight but not resilient to the volume and sophistication of modern ICS data. Their high rates of false positives also promised to overwhelm operators, prefiguring the very alert fatigue that motivates severity scoring today.

3.1.2 Deep Learning Approaches

Deep learning also brought forth models better adapted to ICS data complexity. LSTM networks gained fame in extracting long-range time patterns, and Macas & Wu (2019) demonstrated they were better than OC-SVM on the SWaT dataset. Nevertheless, training was sluggish and hyperparameter-sensitive in LSTMs. Other models were also investigated: CNNs view ICS data as unidimensional time series and can extract local dependencies at lower resource cost (Ahmadzadeh et al., 2024). Autoencoders (AE) and Variational Autoencoders (VAE) identify anomalies based on reconstruction errors, and Niu et al. (2020) used combinations of LSTMs and VAEs to outperform these, although at increased computational overhead. GANs, as in MAD-GAN (Li et al., 2019), created synthetic data to increase robustness, although stability and excessive computation restrained their popularity.

These methods constituted an advancement in nonlinear, multivariable ICS data detectability (Han et al., 2019). The methods, though, carried limitations in computational intenseness, sluggishness in training, and non-interpretability. In live ICS systems, whose operators cherish urgency and confidence in results, these limitations reduce their usability while they boost accuracy.

3.1.3 Lightweight and Unsupervised Models

Realising the limitations of deep learning, researchers moved on to lightweight, unsupervised approaches which balance performance and tractability. Empirical Cumulative Distribution Outlier Detection (ECOD) of (Li et al., 2019) takes advantage of statistical distributions rather than complex network structures. ECOD achieved outstanding performance on numerous benchmarks, and its advantage of being parameter-free and having low computation.

Deep Support Vector Data Description (DeepSVDD) by (Ruff et al., 2018) provided a middle ground among shallow techniques and deep models. It projects the input features onto a hypersphere and assigns points outside the hypersphere as outliers. WaXAI used ECOD and

DeepSVDD on the SWaT dataset and demonstrated that both models had competitive rates of detection, ECOD surpassing in efficiency while DeepSVDD provided better representation learning. These models collectively pointed to a movement toward solutions that are effective and frugal at the same time.

The WaXAI framework used ECOD and DeepSVDD on the SWaT dataset and showed that both models reached competitive detection rates. ECOD was especially efficient, and DeepSVDD shone for its capability of learning complicated feature representations. The combination of these models indicated a trend in the literature toward anomaly detection solutions that are not only efficient but also resource aware.

Regardless of promise, lightweight approaches still produce anomalies as either scalar error scores or as binary classifications. They do not differentiate subtle anomalies of trivial impact from more serious and immediate intervention-worthy anomalies (Bosman et al., 2013). A limitation of such approaches makes more salient the necessity of a severity-scoring system for converting detection outcomes into prioritized, intervention-worthy responses.

3.2 Explainable AI in ICS

3.2.1 The Need for Interpretability

Detection accuracy is most important interpretability is also critical in ICS. Operators must understand why a model has flagged an anomaly before they decide on whether they should disable equipment, send alarms, or continue running. Black-box deep learning models, though accurate, provide no such explanation, and distrust follows with possible overaction or indecisiveness. Explainable AI (XAI) has thereby become the key to anomaly detection in ICS, attempting to inject transparency into machine learning decision making (Emrullah ŞAHİN et al., 2024).

3.2.2 Techniques for Interpretability

Certain XAI methodologies have been applied to ICS. SHAP (Shapley Additive Explanations), from cooperative game theory, returns feature importance values and, thus, presents how each of the sensors or actuators participated toward an anomaly score. LIME (Local Interpretable Model-Agnostic Explanations) produces surrogate models of specific predictions such that local interpretability results. Accumulated Local Effects (ALE) returns mean feature contributions by dampening correlation error inherent in Partial Dependence Plots. Integrated

Gradients (IG) apply gradient-based methods for assigning importance in neural networks (Vermeire et al., 2021).

WaXAI's experimentation revealed how the methodologies perform. SHAP provided the best localization of anomalous elements, though it required extensive computations, which were even higher for ECOD. LIME produced results worthy of explanation, but it was limited by high runtime and not suitable to be labelled as a real-time application (in time-sensitive settings). ALE was quicker, but did not provide as fine of localization, and IG performed well for DeepSVDD, but was not translatable to probabilistic models like ECOD.

3.3 Severity and Risk Assessment Frameworks

3.3.1 IT Security and CVSS

In information technology, severity ranking is commonplace. The Common Vulnerability Scoring System measures software vulnerabilities by assembling exploitability measures (like attack vector and complexity) and impact measures (on confidentiality, integrity, and availability). The result is a numerical score, ranging from low, to medium, to high, to critical, that groups vulnerabilities. CVSS has gained widespread acceptance and drives patch management and incident response.

However, CVSS is not suited well for ICS. CVSS was created for information systems whose impacts are not physical but digital. Anomalies in an ICS may have the potential to endanger human life, cause environmental damage, or break entire supply lines. The confidentiality metrics is less important in ICS, while factors like operational disruption and safety criticality are absent from CVSS. Application of IT severity frameworks to ICS hazards provides an inaccurate reflection of actual criticality of anomalies (Howland, 2021).

3.3.2 Risk Assessment in ICS

ICS work has attempted to extend models of risk assessment, but the vast majority of these are checklist-based and static. A few have looked at attack likelihood modelling, business impact analysis, and high-level risk score. While useful for policy and audit, these models do not integrate real-time inputs from anomaly detection. The models lack the detail and real-time capability needed in the face of an active attack (Kim et al., 2022).

Other studies have integrated anomaly detection with explanation. Anomaly detection with feature selection was investigated by Roshan and Zafar (Roshan & Zafar, 2022) and involved

connecting importance scores to criticality by proxy. Yet, no attempt was made in any of these works at providing a systematic approach to estimating severity at the process level. The gap stands stark: although detection and explanation have progressed significantly, severity scoring has not been studied to a great degree for ICS security.

3.3.3 The Gap and Research Opportunity

The absence of operational dynamic severity score approaches is the major caveat in ICS security research. Anomaly detectors produce alarms, and XAI produces explanations, but operators at industry plants make urgency decisions based on personal judgment, nonetheless. The consequence is inconsistency and delay in incident response. Therefore, there is a need for a severity scoring framework that can assist the operators in prioritising their actions in case of a cyber-attack.

The challenge is to devise a scoring methodology that acknowledges the unique nature of ICS. This includes discerning the safety critical nature of certain elements (i.e., pumps and chemical dosing systems), the value of each stage of process flow (i.e., water intake and chemical treatment), and the level of operational disruption introduced by deviations. By synthesising these elements, severity score and detection can be conveyed from a Boolean indication to an advanced decision support system.

Chapter 4

Methodology and Implementation

This chapter provides an instance of the severity scoring framework constructed in this dissertation, expanding upon WaXAI (Mathuros et al., 2024). WaXAI showed an example of utilizing unsupervised models of anomaly detection such as ECOD and DeepSVDD in Industrial Control Systems (ICS) and making their outputs explainable by means of Explainable AI methods. It, however, did not allow measurement of how far-reaching these anomalies were, and as such, operators could not obtain any guidance regarding prioritisation. The primary contribution of this dissertation is thus the conceptualisation of a severity scoring framework assessing these findings in three dimensions: safety criticality of the part, stage significance in the treatment process, and operational interruption during attack. This transforms raw anomaly finds into prioritised scores upon which operators can act. The rest of this chapter captures the dataset and preprocessing (Section 4.2), in-depth details on the models ECOD and DeepSVDD (Sections 4.3–4.4), XAI's role (Section 4.5), framework for severity design (Section 4.6), validation thereof (Section 4.7), operator-centric notifications (Section 4.8), case study (Section 4.9), and lastly challenges and limitations (Section 4.10).

4.1 Implementation Workflow

The implementation workflow summarized in Figure 2 begins with dataset preprocessing of the SWaT dataset to guarantee that we have input data that is clean and structured for training and evaluating the model. The pre-processed data is then passed onto the anomaly detection model, where we employ two models: ECOD (the probability-based baseline) and DeepSVDD (the detector based on of neural representation), to detect deviation from normal behaviour. Subsequently we introduced XAI implementation, to enhance human interpretability of the detections; XAI implementation provided explainability of the detections by allowing the user to examine which features(s) resulted in an anomaly detection. Following we processed the detections with the severity scoring framework, we calculate scores across three dimensions, being safety criticality, stage importance, and operational disruption. Continuous scores were classified into Low, Medium, or High/ Severe for scoring severity. To assess the framework, we compare predictions against an expert defined ground truth mapping, with benchmark severity labels for each attack mandate / supporting the expert centre benchmark labels against MCDA generated severity labels. Finally, we reached the last stage operator

notification/evidence-based reporting phase, which creates structured alerts/reports for operator decision making in the real-time industrial control system space.

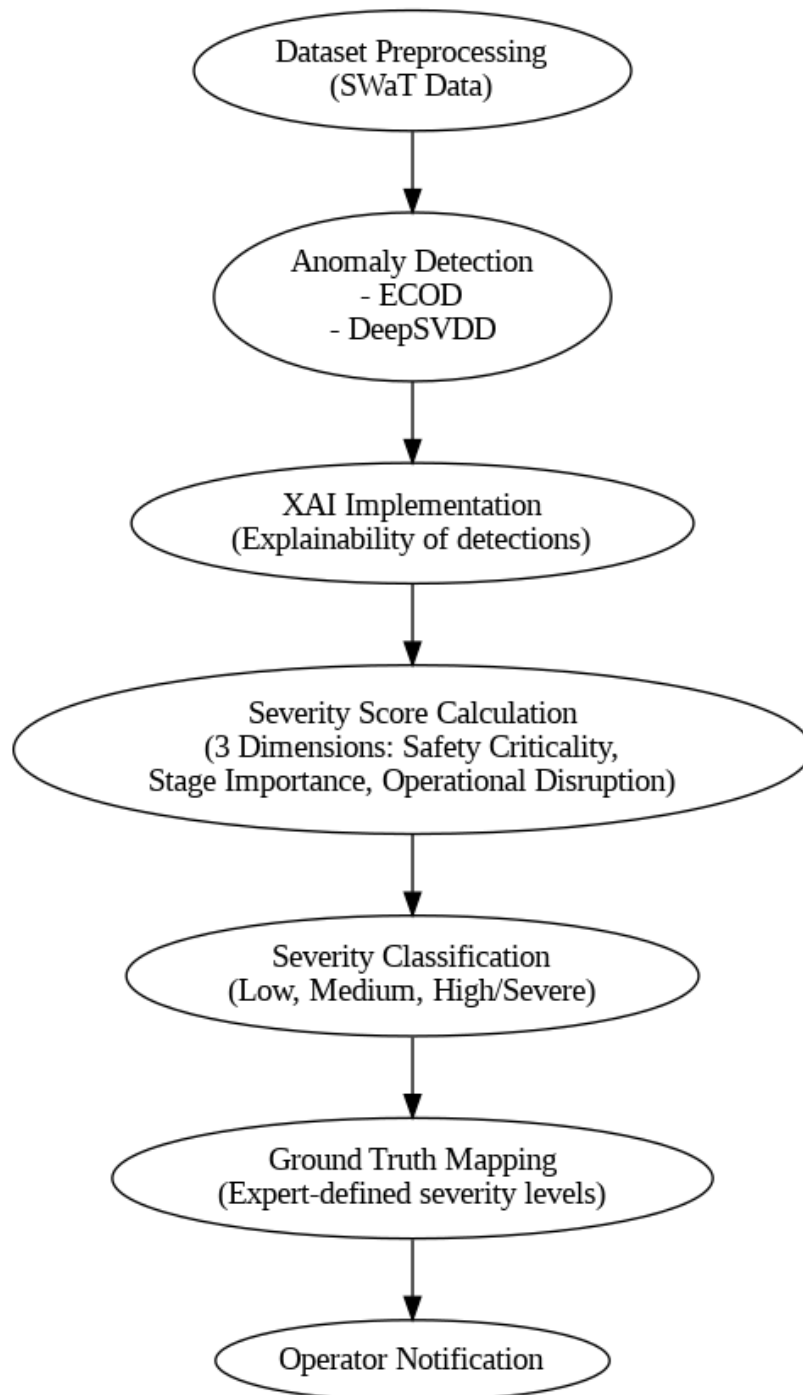


Figure 1: flowchart of workflow

4.2 Dataset and Preprocessing

The dataset that was chosen in this application is the Secure Water Treatment (SWaT) dataset, being one of the most widely used benchmarks in ICS security research. SWaT is the scaled but functional water treatment plant installed at the Singapore University of Technology and design that mimics the six-stage purification process. Along the stages, there are the 51 sensors and actuators continuously logging values related to the flow, pressure, chemical concentration, tank levels, and status of valves or pumps (Goh et al., 2017).

The data contain 11 consecutive days of operations. The initial seven contain normal operations, including steady-state plant behaviour with no intervention. The last four contain the release of 41 staged cyberattacks that were intentionally constructed to be like realistic adversarial activity against water treatment assets. Each assault is aimed at target sensors or actuators, in this case, through tampering with readings, imposing component states, or injecting anomalous control logic. Note that the data is accompanied with a ground-truth file that captures the inception and end times of each attack, along with the attacked component(s).

Preprocessing then followed the procedure outlined in WaXAI. Raw timestamp was converted into standard datetime type so that direct alignment with ground-truth attack intervals was possible. Sensor measurements that run-in units (litres per second, percentage valve openings, bar pressure, etc.) were normalised in the $[0,1]$ range so that they can be compared across heterogeneous features. The data was then separated into training data (normal operations only) and test data (normal and attacks).

This dissertation reuses the WaXAI's preprocessing pipeline so that the results become instantly comparable. The benefit of my work is not in re-implementing the preprocessing but in enriching the outputs of the detection of the anomalies with scores of their severities.

The six-stage SWaT water treatment testbed is displayed in Figure 1, which should provide a general orientation of the configuration of all sensors and actors involved in the process. Table 1 depicts how devices are distributed throughout each stage, mapping actuators (pumps/valves) and analysers (sensors) to their role in operation.

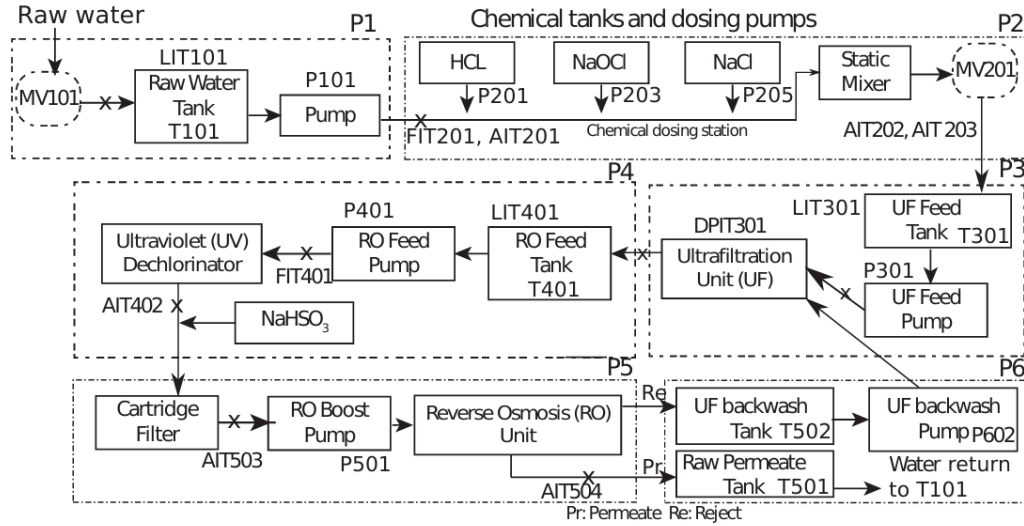


Figure 2: Overview of SWaT testbed (Mathur & Tippenhauer, 2016)

Process	Sensors	Actuators
P1	LIT-101, FIT-101	MV-101, P-101
P2	AIT-201, AIT-202, AIT-203, FIT-201	MV-201, P-201, P-202, P-203, P-204, P-205, P-206
P3	DPIT-301, FIT-301, LIT-301	MV-301, MV-302, MV-303, MV-304, P-301, P-302
P4	AIT-401, AIT-402, FIT-401, LIT-401	P-401, P-402, P-403, P-404, UV-401
P5	AIT-501, AIT-502, AIT-503, AIT-504, FIT-501, FIT-502, FIT-503, FIT-504, PIT-501, PIT-502, PIT-503	P-501, P-502
P6	FIT-601	P-601, P-602, P-603

Table 1: Sensor and actuators in each process

Note: Device codes (e.g., LIT101, MV202, P602) follow the SWaT testbed convention. The first three letters indicate the device type (LIT = Level Indicator Transmitter, MV = Motorized Valve, etc.), and the last three digits specify the stage (first digit) and the device number within

that stage. Table 2 explains the naming convention for devices in specificity of the abbreviations and functions so that any attack point or anomaly analysis can be interpreted consistently.

Device Code	Full Form / Description	Function in Plant
LIT-XXX	Level Indicator and Transmitter	Measures tank water levels.
FIT-XXX	Flow Indicator Transmitter (Flowmeter)	Measures water flow rate.
AIT-XXX	Analyser Indicator Transmitter	Monitors chemical properties.
DPIT-XXX	Differential Pressure Indicator Transmitter	Measures pressure difference across filters/pipes.
P-XXX	Pump	Moves water/chemicals through treatment stages.
MV-XXX	Motorized Valve	Controls water flow between tanks/pipes.
UV-XXX	Ultraviolet DE chlorinator	Performs water disinfection.
PIT-XXX	Pressure Indicator Transmitter	Measures system pressure.

Table 2:SWaT Device Naming Convention

4.3 Probability-Based Model: ECOD

The initial algorithm utilized in WaXAI was Empirical Cumulative Distribution Outlier Detection (ECOD). ECOD is a parameter-free algorithm based on the approximating probability distributions that estimates the probability of observations utilizing the empirical cumulative distribution function and marks the observations in the tails of the distribution as being anomalous.

ECOD was chosen due to two fundamental reasons. First, ECOD is computationally efficient and can be scaled up to large data sets, critical in ICS applications that produce high-rate, multivariate data streams. Second, no parameter tuning is required, which renders ECOD especially desirable in operational environments where ease of deployment is critical (Li et al., 2022).

In WaXAI, the ECOD was learned from the normal SWaT data of the seven days and then tested against the complete dataset. The anomalies were detected based on a threshold at the 95th percentile of the scores indicating anomalies with the addition of a 100-second persistence rule to eliminate transient variations that would be incorrectly labelled as anomalies.

In this dissertation, ECOD was left as the probability-based baseline detector. Nevertheless, the anomaly scores supplied by ECOD were not used directly in the severity measure. The framework instead emphasized the attacks identified by ECOD (after thresholding and persistence filtering). For each identified attack, the three novel dimensions (safety criticality, stage importance, and disruption in operations) were then used in computing the severity scores. ECOD in this manner acted as the gateway into the framework of severity scoring based on the idea that the novel contribution of this dissertation could be run over the identified set of attacks.

4.4 Neural Model: DeepSVDD

The other anomaly detection model that was used in WaXAI was Deep Support Vector Data Description (DeepSVDD), which generalizes the classical one-class SVM formulation using neural representation learning. DeepSVDD converts the input data in the latent space based on the neural network and learns the hypersphere that covers the bulk of normal training data. The data points that appear in the space outside the hypersphere are considered as anomalies (Jiang et al., 2023).

DeepSVDD also has some strengths over superficial models. Using neural networks, DeepSVDD can detect nonlinear and high-order dependencies among variables in multivariate ICS data. This is appropriate in the case of systems like SWaT, whose behaviour is due in part to nonlinear interaction among pumps, valves, and sensors. These strengths, however, come with the drawback of being more computationally intensive. DeepSVDD entails longer training times up to 100 epochs in the case of WaXAI—and is enhanced using the GPU, but this latter is often unavailable in the field.

In WaXAI, DeepSVDD was trained on normal SWaT data during the first seven days and then validated on the complete dataset with normal working and attacks. The anomalies were detected with the guidance of the 95th percentile threshold over the anomaly scores, as well as the 100-second rule of persistence to dampen false alarms from short-term fluctuations. In this

thesis, DeepSVDD was employed as the neural-based baseline detector in combination with the probability-based ECOD. Just as with ECOD, the raw anomaly scores themselves were not included in the formula for severity scores. Instead, DeepSVDD was responsible only for identifying which attacks were detected. For each confirmed attack identified by DeepSVDD (after thresholding and persistence filtering), the proposed framework of safety criticality, importance of stage, and disruption of operations was used to calculate severity scores. In this manner, DeepSVDD added value to the pipeline in the form of an alternative detection path, while the framework of severity scoring added extension of its outputs in the form of prioritisations in support of decision-making.

4.5 Explainable AI Integration

In response to the black-box issue of anomaly detection, WaXAI included four explainability (XAI) techniques: SHAP, LIME, ALE, and Integrated Gradients. Each technique includes feature attributions to help operators understand which variables contributed to the anomaly decision. WaXAI found that SHAP provided the most reliable localisation of attacked components, though at substantial expense; LIME was slow and essentially produced unreliable attributions over time; ALE was fast and yielded global insights but lacked individual precision; and Integrated Gradients works well with neuro-models but not with probability-based models.

While the consideration of XAI about operator trust is acknowledged, this dissertation focuses on severity scoring rather than interpretability. WaXAI used XAI to identify impacted components. This dissertation takes a different dimension, operational disruption, which captures how much of a disturbance to a process occurs from the operational perspective, to effectively guard against the black-box nature of anomaly detection. Therefore, the severity scoring framework is complementary in relation to WaXAI's XAI outputs but separate in terms to how it is used.

4.6 Severity Scoring Framework (Novel Contribution)

4.6.1 Motivation

Anomaly detection in ICS is worthwhile but inadequate in practical application. The operators could be presented with numerous alerts at the same time, with each of the alerts differing in

importance. If all the anomalies are considered with equal importance, then there would be inefficacy as well as more seriously, the possibility of missing critical attacks due to trivial disturbances. For instance, the slight variation in one of the redundant flow meters will perhaps not demand instant correction, but the long-term manipulation of one of the dosing pumps will instantly threaten human safety.

Severity scoring system bridges the gap through the augmentation of WaXAI with a systematic way of measuring the detected anomalies in terms of severity. The anomalies are assessed along the lines of safety criticality, stage importance, and disruption in operations. These three dimensions collectively offer an elaborate score that covers both the technical as well as the operational impacts of the anomalies.

4.6.2 Safety criticality of Components

The first dimension of the severity scoring framework assesses the safety criticality of the components affected during an anomaly. Not all components in the SWaT plant are equally hazardous when an anomaly involves them. Some components affect the safety of drinking water or human life, while others mainly affect the lifespan of equipment or service/device quality. Instead of assigning equal weights to all components from a safety perspective, I classified components into three criticality categories: L1 (high criticality), L2 (medium criticality), and L3 (low criticality) with weights of 1.0, 0.7, and 0.2 respectively. Here, L indicates a level of Loss Impact based on the safety criticality: L1 is the highest loss impact (direct impacts to human safety; or to water quality), L2 is medium loss impact (damaging the equipment or losing money), and L3 is low loss impact (affecting only service performance or the quality of monitoring).

- **L1 – High Critical (Weight = 1.0):**

These components are directly related to human intervention or the safety of water. Any failure or manipulation can put public health at risk or cause catastrophic failure of the processes. Pumps (P-101 raw water intake pump, P- 102 main process pump) allow for flow of water continuously and if they fail, there will either be no raw water supplied or the whole process would stop flowing. MV- 101 is the main motorised valve which directly is used to help gauge raw water inflow; without it the system would not be stable. LIT-101 evaluates what level the raw water tank is to make sure they don't overflow, allowing for the pumps to run dry, creating an unsafe condition. P-602 is the reverse osmosis (RO) high pressure pump which is used for the final stage of the purification of the water, if there are

any anomalies, water quality cannot be guaranteed. By assigning these components the highest weight it ensures that any attacks on them must always take precedence over others.

- **L2 – Medium Critical (Weight = 0.7):**

Components in this category have a risk of poor equipment condition (and/or financial losses) through compromise. Although if these pieces do fail, they do not directly put human life in jeopardy, they may significantly degrade the integrity of the process, increase the operational costs or lead to issues down the process. Examples here are ultrafiltration valves MV-302, MV-303, MV-304 and then differential pressure transmitters DPIT-301, in a position to limit or minimize ultrafiltration itself. Sustained attacks to these instruments can lead to membrane damages and/or costly repair.

- **L3 – Low Critical (Weight = 0.2):**

For the remainder, most components will be classified as L3 since losses, or compromise of the device, only affects performance of the service or process, and not their safety. Flow transmitters or auxiliary pumps; or other analysers and redundant sensors (e.g. FIT-401, AIT-301, P-401, UV-401), would fall into the L3 category. These present the least trouble, because an attack would cause inaccurate reporting and/or degraded optimisation, while the core process could be maintaining a safe operational state. They are less impactful than IT and OT L1 and L2 devices.

The remaining, majority of components apply to L3 because a loss or compromise only affects service or process performance and not safety. Flow transmitters, auxiliary pumps, more analysers and/or sensors that are redundant (e.g. FIT-401, AIT-301, P-401, UV-401) fall into L3 category. These devices are less of an issue because an attack can result in incorrect reporting and/or degradation of optimisation, but the core process can continue running safely. These devices have less impact than L1 and L2 devices.

Criticality Level	Weight	Components
L1 – High Criticality	1.0	P-102, MV-101, LIT-101, P-101, P-602
L2 – Medium Criticality	0.7	MV-304, MV-303, DPIT-301, MV-302, MV-201, LIT-401, LIT-301, FIT-601, AIT-501.
L3 – Low Criticality	0.2	FIT101; AIT101; AIT102; AIT103; MV301; AIT301; AIT302; AIT303; P301; P302; AIT401; AIT402; FIT401; UV401; P401; P402; P403; P404; AIT503; AIT504; FIT501; FIT502; FIT503; FIT504; P501; P502; PIT501; PIT502; PIT503; AIT601; AIT602; AIT603; P601; P603; PIT602; PIT603; AIT202; MV504; P203; P201; P205; AIT-502; AIT502

Table 3: Component Safety Criticality Mapping

4.6.3 Stage Importance

The second degree of significance in the severity framework recognizes the significance of the operational stage where the anomaly occurs. Although all six operational stages of the SWaT water treatment process contribute to plant operation, the disruption of different operational stages of the water treatment process do not all have the same level of risk. Some operational stages have direct implications on water quality and safety, while the majority restrict operational efficiency, or disposal related to water treatment. For this reason, it was decided to assign a weight between 0 and 1 for each operational stage, with a weight closer to 1 reflecting the greater significance of that operational stage with respect to overall plant output.

- P1 – Raw Water Intake (Weight = 0.8):

The raw water intake stage takes responsibility for extracting water into the treatment process. If anomalies happen in this stage, this can impact the whole downstream processes by starving the downstream stages of water or possibly overflowing at the source. Compromise of the pumps or valves in this stage will not directly contaminate the quality of the water but may occur with extreme loss of operational control to whole large operational areas. Therefore, P1 has a high weighting of 0.8.

- P2 – Chemical Dosing (Weight = 0.6):
The chemical dosing stage maintains injections of the coagulant dosing and other chemical dosing required to neutralise impurities present and leave the water in an appropriate state for further treatment. Anomalies associated with P2 could disrupt the chemical balance of the water, and in turn the efficiency of the treatment. However, effects from an impaired chemical dosing stage may be generally addressed in later processes, such as ultrafiltration or disinfection stage. For this reason, P2 is weighted at 0.6, below intake and disinfection stage for importance in the water treatment process.
- P4 – UV De-chlorination (Weight = 0.7):
This step is not for disinfection of microorganisms (that was accomplished previously with chemical dosing and again in the chlorination). But in SWaT, the UV chamber is utilized to remove free chlorine from the supply water before it reaches the RO membranes since they would be destroyed if they are exposed to chlorine. Sodium bisulphite is occasionally fed here too to control oxidation-reduction potential (ORP). Anomalies here do not immediately precipitate microbial risk, but they can destroy the RO membranes, which are responsible for final purification. Therefore, P4's score is a high 0.7: not as plant-shutting as P1, but interferences here immediately compromise downstream treatment's integrity.
- P3 - Ultrafiltration (weight = 0.5):
Ultrafiltration removes very fine particles from the feed water as well as any other impurities. A failure in this stage only creates bad water quality for the treatment process, and only in the case of an extreme failure would there be damage to the equipment. If the treatment stage is bypassed or breaks down, there is some residual defence from the following stages in the treatment process, disinfection and polishing. Because there might still be some protections for water quality even with a failure in the ultrafiltration stage, P3 is weighted a moderate score of 0.5.
- P5 – Reverse Osmosis (Weight: 0.3):
RO is the final polishing step of the treatment process. It eliminates dissolved salts and other micron impurities and results in high-quality permeate water. It is less safety-critical compared to previous steps, however, since water going to this step is already disinfected, filtered, and dechlorinated. An RO fault generally affects water quality (taste, clarity, or mineral balance) rather than immediately affecting safety. It is thus weighted less (0.3), indicative of its value for quality assurance rather than first-line health protection.
- P6 - Drainage or Disposal (0.4 weights):

The last stage of treatment is the drainage or disposing stage. If there is an anomaly here, it may contribute to environment contamination (e.g., effluence of untreated water) or flushing system operational problems and does not relate to safety of drinking water. As these kinds of anomalies affect only the water flushed, P6 is assigned the second lowest value of 0.4.

This classification system represents a balance between safety critical operations (e.g. P4 disinfecting); system-wide dependencies (e.g. P1 intake); and ancillary activities (e.g. P5 polishing, P6 disposal). The weighted calculations allow any anomaly in earlier or safety dependent cases to predominate the severity calculations, while recognizing downstream impacts.

4.6.4 Operational Disruption

The third dimension of the severity scoring framework is operational disruption, which also corresponds to the amount of disruption to the normal operations of the water treatment plant. Whereas safety criticality and stage significant are structural and static properties of the system, operational disruption is only based on the behaviour of the plant at specific time windows. The operational disruption dimension shows not only what is impacted, but by how much or to what extent it is impacted.

We measured operational disruption by percentage deviation of the nominal operating range of impacted actuators and sensors. There were three key steps involved in the measurement of disruption:

- **Baseline Estimation:**

For each of the components, a baseline was established by taking its median nominal value across all regular (non-attack) time segments of the SWaT dataset. The use of the median over the mean was chosen given its resilience to noise and fluctuations of benign sensor data and then acting as a sensible baseline for comparisons.

- **Deviation Calculation:**

During an attack, the deviation of each component was calculated at every time step using the following formula:

$$\text{Deviation}(t) = |\text{Observed}(t) - \text{Median Nominal}| / |\text{Median Nominal}|$$

This formulation describes the extent to which the observed value indicates deviation from the expected nominal behaviour. The normalisation by the median nominal value ensures that all sensors and actuators are cross-comparable, operating in different scales.

- **Maximum Deviation and Ranking:**

For each shaken component we assessed the maximum deviation record through the attack window to encapsulate the most dramatic shaking during the attack. After calculating the maximum deviation respective of and for all the components, the selections were sorted from most to least disturbed, top 10 disturbed components were established, and these were selected as the main drivers of disturbance. To help reduce the consideration of extreme outliers, we took the median deviation score of the top 10 components to be the scalable operational disruption scores useful for the attack. The operational disruption score or raw score was normalized to a set range for comparability of the last two severity levels and were thereafter combined into the overall severity formula.

This design has three elements. One, the framework is robust against natural sensor variability as it establishes a baseline against median nominal values. Two, though it captures the Top-10 most disrupted components, by using the median deviation, it requires that the variability on a single extreme sensor does not drive an overestimation, while also not underestimating opportunity loss by averaging over all components. Three, normalizing its scores means it can be proportionally integrated against the criticality of safety and importance of the stage in the final constructed severity score and ensures that operational disruption does not solely drive the severity score by virtue of scale differences.

This dimension therefore captures the dynamic footprint of attacks on plant behaviour. Think about it: a brief yet high-magnitude spike from a single flow sensor may only lead to a moderate disruption score, however, a sustained deviation from multiple pumps and valves will reflect a high disruption score. In this manner, operational disruption, rounds out safety criticality and stage importance and ensures the severity scoring reflects both structural importance and observed impact.

4.6.5 Severity Scoring Formula

The three dimensions were combined into a severity formula using the below structure

$$\text{Severity Score} = \alpha \times \text{Safety Criticality} + \beta \times \text{Stage Importance} + \gamma \times \text{Operational Disruption}$$

In this formula, α , β , and γ act as weighting factors, showing the relative contribution of safety criticality, stage importance, and operational disruption to the overall score. In this implementation, the weights were set as 0.5, 0.25, and 0.25 respectively, which allowed it to put slightly more emphasis on safety criticality, while considering process stage and disruption as also being an important consideration. These values were chosen because they were representative of the dissertation's safety-first motivation but could be fined-tuned based on expert opinion or future studies.

The severity scores created were continuous values between 0 up to 1. Thresholds were specified to create categories of low (<0.45), medium ($0.45 < 0.75$), high (>0.75) severity levels of scores.

In cases where some components' median nominal value is close to zero (such as flow sensors with the valve closed), a small epsilon correction ($\epsilon = 1e-6$) was added to the denominator to avoid distortion. This added stability to the calculations and mitigated potential division-by-zero errors and extreme deviation scores.

This formula allowed for the anomaly detections to be transformed into structuralised, prioritised scores that operators could interpret in a consistent manner.

4.6.6 Edge Cases

In applied settings, anomaly detection and severity scoring systems need to have edge cases taken into consideration. Edge cases exist when standard operating procedures could yield misleading and/or sporadic outcomes, if they are not handled correctly. Numerous barriers were put in place throughout this dissertation to guarantee the output of the severity framework was interpretable, and consistent as possible, If it was put in an unusual situation.

1. Missing Values and Noisy Data

ICS datasets like SWaT, occasionally have missing values or sensor noise. Communication delays, sensor drift due to calibration inconsistency, or errors in logging can produce noisy data (Maged, 2022). If noisy data was treated as actual disruptions, the severity score would reflect false positives. Before baseline medians were created, median from clean and non-attack

segments only were used. If the data was considered noisy, as part of preprocessing, it was eliminated completely as an outlier if it did not confine to a possible attack pattern. This ensured deviations were always based on a stable and representative baselines.

2. Very Short Attacks (<100 seconds)

Transient deviations can appear as spike and may not be indicative of consistent malicious behaviour. In this sense, applying the 100 second persistence rule to these instances (adopted from WaXAI), would return the anomaly as a transient spike, and not classify them as malicious events. Therefore, the operator would not be overwhelmed and fatigued by false alarms. As explained in chapter 1, if the anomaly persisted longer than 100 seconds, the alarm would remain active as the attack would be classified as attack with the persistence rule applied.

3. Zero or Near-Zero Baseline Values

When certain sensors have a median nominal value that is close to zero (e.g. flow when a valve is closed) dividing by these small nominal values could exaggerate the deviation value. In these instances, a small epsilon correction factor was implemented in the denominator to ensure there was no zero/near-zero division error and to avoid unreasonable deviation scores.

4. Extreme Outliers

Occasionally, for a single sensor, there is extreme deviation in one sensor while the rest of the sensors fail to show significant deviation. Relying solely on deviation from one sensor with maximum deviation would likely create an exaggerated severity of an attack. Thus, the framework has relied on taking the median of the Top 10 components and not relying on the maximum. This design consideration was to achieve an adequate balance between disruption sensitivity/resilience toward extreme outliers.

5. Multi-Stage Involvement

Some incidents will influence components that cross multiple stages. In these instances, severity scoring must avoid a bias toward the highest stage or to the lowest stage. This framework takes the average of the stage importance of all affected components, to provide a severity data point that reflects the multi-stage nature of these kinds of incidents and does not inaccurately elevate the importance of a higher stage or depress the severity of a lower stage.

Discussion of Edge Case Treatment

By explicitly planning for these situations, the severity framework provides robustness in actual implementation. Edge case treatment ensures that misleading results do not occur when raw calculations could otherwise inflate, diminish, or skew severity scores. For operators, this means that the system will allow for prioritisation and allow for prioritisation while being confident that ambiguity of data, minor disruptions, and dimensional distortion should not be debated as to severity.

4.7 Validation Against Expert-Informed Ground Truth

The Secure Water Treatment (SWaT) dataset provides detailed logs of 41 staged cyberattacks on an operating water treatment testbed; including times the attacks started and ended, including which components were manipulated. However, the SWaT dataset does not provide graded severity levels for the staged attacks, it does provide a binary label of attack, or no attack. In our view, this is only sufficient for benchmarking anomaly detection methods, but it is insufficient for evaluating a severity scoring framework which is to not just detect anomalies but rank them by operational criticality.

An author-directed, ground truth severity mapping was created with the approval of the project research supervisor. This mapping reflects the author's extensive review of all 41-case study staged attacks in the SWaT dataset by considering their impact on plant safety, equipment, and water quality. Each attack level was assigned to three severity levels, as follows:

- Low severity - minor disturbances which fundamentally targeted non-critical sensors or redundant monitoring sensors/objects. This means that the attacks led to degradation in service quality but did not compromise equipment or water safety.
- Medium severity - involve attacks that could have led to compromising the integrity of the process or the integrity of the water quality, if adequate corrective actions were not taken. Some of these attacks include some of the valves or level sensors in order to embryo ideal conditions during mid-critical pathway stages of the drinking water treatment.
- High/Severe severity: Attacks could cause significant equipment damage, cause a complete shutdown of the process, or directly risk safety.

This mapping serves as the evaluative baseline for consideration of the severity scoring framework presented in this dissertation.

Ground truth severity table

The full mapping of the 41 SWaT attacks is shown below.

Attack #	Primary Component(s)	Ground Truth Severity
1	MV-101	Medium
2	P-102	High/Severe
3	LIT-101	Low
4	MV-504	Low
6	AIT-202	Low
7	LIT-301	Low
8	DPIT-301	Low
10	FIT-401	Low
11	FIT-401	Low
13	MV-304	Low
14	MV-303	Low
16	LIT-301	Low
17	MV-303	Medium
19	AIT-504	Low
20	AIT-504	Low
21	MV-101 + LIT-101	Medium

Attack #	Primary Component(s)	Ground Truth Severity
22	UV-401, AIT-502, P-501	Low
23	P-602, DIT-301, MV-302	Medium
24	P-203, P-205	Low
25	LIT-401, P-401	Low
26	P-101, LIT-301	Low
27	P-302, LIT-401	Low
28	P-302	Low
29	P-201, P-203, P-205	Low
30	LIT-101, P-101, MV-201	Medium
31	LIT-401	Medium
32	LIT-301	Medium
33	LIT-101	Medium
34	P-101	Low
35	P-101, P-102	Medium
36	LIT-101	Low
37	P-501, FIT-502	Low
38	AIT-402, AIT-502	Low
39	FIT-401, AIT-502	Low
40	FIT-401	Low

Attack #	Primary Component(s)	Ground Truth Severity
41	LIT-301	Low

Table 4: ground truth mapping of the attacks

4.8 Notification and Reporting

To put severity scores into practice, the project developed an organized reporting system. Instead of delivering scores from anomaly detection to operators for evaluation, the system created written notifications.

For example:

“High-severity anomaly detected in P-102 (Stage 2: Chemical Dosing). Recommended Action: Immediate operator intervention required.”

Notifications like these are short, easy to understand, and directly applicable to the real-time feedback and decisions required of ICS operators. They integrate sophisticated machine learning outputs and operational messaging that existed inside control rooms.

Chapter 5

Results

This chapter presents and examines the results of the proposed severity scoring framework, applied to the SWaT dataset, based on the anomaly detections of the ECOD and DeepSVDD models. The objective of this performance evaluation is to assess whether the framework successfully converts anomaly detections, into severity classifications, in a meaningful way, that could provide operators with prioritised decision-making support.

5.1 Results for ECOD (Probability-Based Baseline)

5.1.1 Overview of model results

The empirical cumulative distribution outlier detection (ECOD) model was trained on seven days of normal data from SWaT and tested against the attack dataset. In this case study, the ECOD model was able to detect 23 attacks, which were then feed into the severity scoring framework for classification. The output returned from this experiment is shown in the confusion matrix and the classification report (Figures 3 and 4). It was observed there was broadly similar detection behaviour to that in the WaXAI paper, however slightly fewer attack detection than the WaXAI study. This small difference in detected attacks is likely related to computational limitations and variations in the runtime configurations, however the general observation for ECOD being a lightweight and a consistent baseline detector for the severity scoring framework is correct.

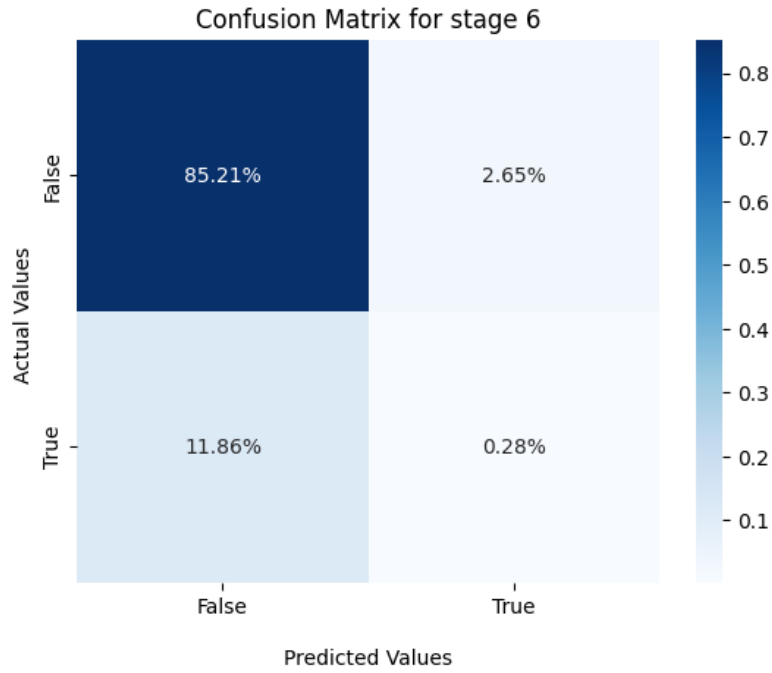


Figure 3: Confusion matrix of ECOD model

	precision	recall	f1-score	support
0	0.05	1.00	0.09	34
1	1.00	0.03	0.06	697
accuracy			0.08	731
macro avg	0.52	0.52	0.08	731
weighted avg	0.96	0.08	0.07	731

Figure 4: Classification report of ECOD model

5.1.2 Confusion matrix interpretation – Severity score

Fig. 5 is a confusion matrix for ECOD-based severity scoring. The rows are based on ground truth by experts, and columns are based on labels from the framework.

- Low Severity (Actual): 11 of 15 ground-truth Low attacks were precisely predicted as Low, and 4 were conservatively promoted to Medium. Not a single Low case was misclassified as being upgraded to a higher level.

- Medium Severity (Actual): Of 7 ground-truth medium attacks, 5 were classified as Medium correctly, and 2 were under-estimated as Low. Not a single case of Medium was misclassified as High.
- High Severity (Actual): There was only a single High attack in this subset. The attack was incorrectly classified as a Medium, so failed to produce it as a high case.

For the most part, there's a strong ability by the framework to distinguish between Low and Medium, although there are two dominant patterns:

- Conservative bias where a few of the Low assaults are upgraded to Medium (false positives in Medium).
- The risk of underestimation, wherein a few of Medium's assaults are estimated as Low (false negatives in Medium).

First, the system produced no false High classifications, and was, as such, stable, avoiding false inflation of severity.

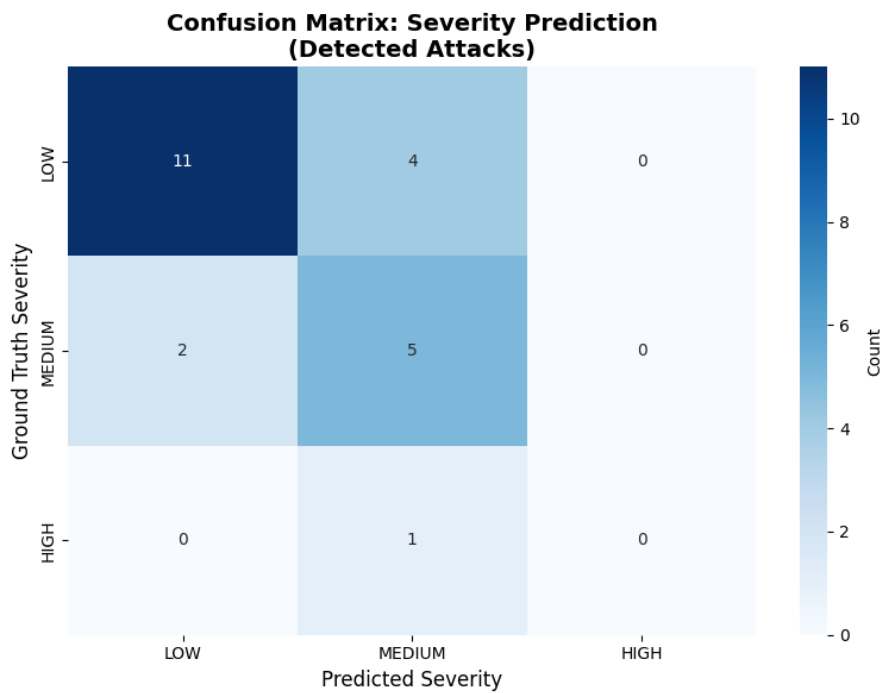


Figure 5: Confusion matrix of ECOD severity analysis

5.1.3 Classification metrics

The classification report shown in Figure 6 shows per-class precision, recall, and F1-scores:

- Low Severity: Precision = 0.85, Recall = 0.73, F1 = 0.79 (Support = 15). The framework was very reliable when predicting Low cases even though a couple of these Low cases were classified improperly as Medium.
- Medium Severity: Precision = 0.50, Recall = 0.71, F1 = 0.59 (Support = 7). The model after the data augmentation approach performed great at capturing the majority of the Medium attacks (very high recall) however several were classified as false positives reducing execution to Medium Severity.
- High Severity: Precision, Recall, F1 = 0.00 (Support = 1). The one recorded High case was not identified correctly.

The overall accuracy was 70% at classifying 16 of the 23 attacks correctly, I also calculated the precision, recall, and F1 scores for the model as an average basis:

- Macro Average: Precision = 0.45, Recall = 0.48, F1 = 0.46 - which represent a lower performance when classes are weighted evenly.
- Weighted Average: Precision = 0.70, Recall = 0.70, F1 = 0.69 - which represent a better performance as a weighted majority for Low and Medium class where Low and Medium attacks were most recorded attacks (support, Low = 15, Medium = 7, High =1).

These metrics demonstrate the framework is very reliable for Low attacks, somewhat reliable for Medium, and not very helpful with respect to High attacks both due to the nature of the class imbalance.

2. CLASSIFICATION REPORT:

	precision	recall	f1-score	support
LOW	0.85	0.73	0.79	15
MEDIUM	0.50	0.71	0.59	7
HIGH	0.00	0.00	0.00	1
accuracy			0.70	23
macro avg	0.45	0.48	0.46	23
weighted avg	0.70	0.70	0.69	23

Figure 6: Classification report for Severity score of ECOD model

5.1.4 Severity distribution – Bar graph

The severity distribution bar chart shows:

- Ground truth contained 15 Low, 7 Medium, and 1 High attack.
- Predictions shifted to 13 Low, 10 Medium, and 0 High.

The bar chart identifies two important behaviour patterns of the framework. First, a few Low-severity attacks were prudently escalated to Medium, in consideration of the safety-first bias that puts the caution of underestimating those anomalies in critical process areas above all else. Second, the one High-severity incident predicted in the ground truth was predicted as Medium, which would show that the median aggregation used does not always amplify the severity of the high-criticality component when all other contextually affected components are exhibiting smaller-than-normal deviations. It is important for us not to mistake a spurious spike as a predicted exceptional score in the context of combined attacks, but this aggregation does provide less sensitivity for rare, catastrophic occurrences. Overall, the system produced an approximately balanced predicted split between Low and Medium (13 v 10), but the lack of predicted High cases demonstrates the influence of both data set imbalance and conservative thresholds.

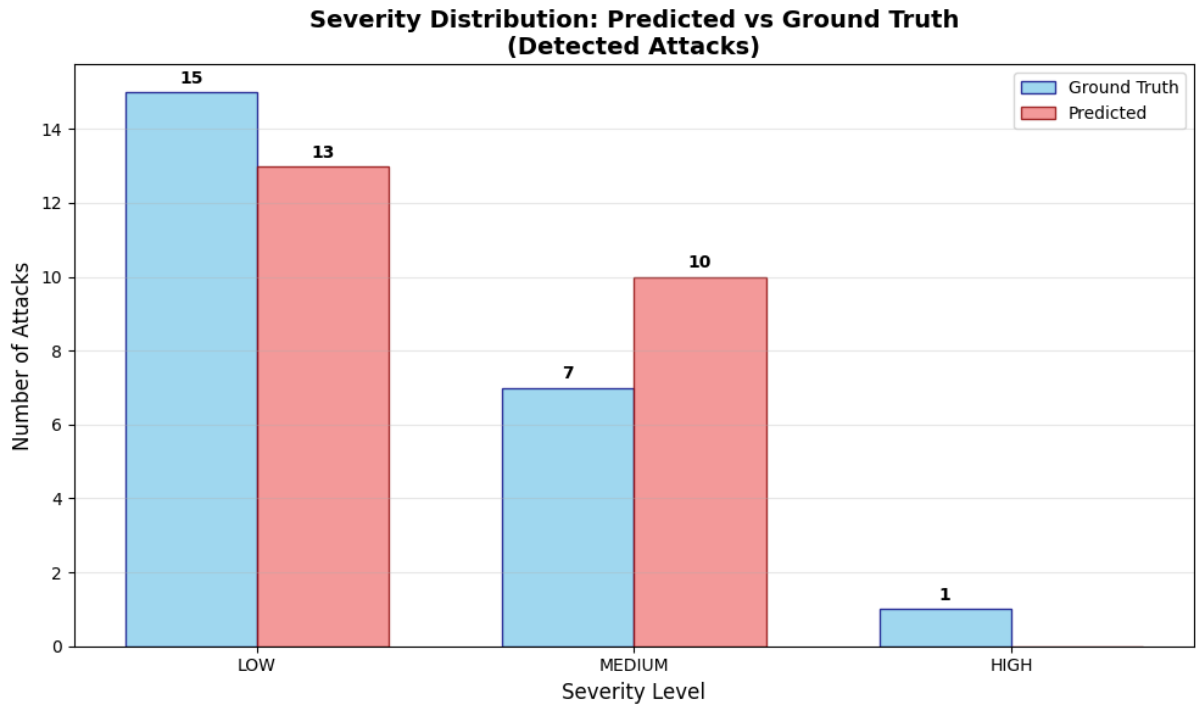


Figure 7: Bar graph of Predicted vs ground truth ECOD model

5.1.5 Stage-wise severity distribution analysis (Heatmap)

Stage-wise heatmap shows the distribution of severity predictions over the six stages of SWaT. Most escalations went to P1 (Raw Water Intake) (4 Low, 6 Medium) given its core role whereby small deviations have downstream spreading. Only a single instance of Low went to P2

(Chemical Dosing) and no escalations, showing limited extent of anomalies. P3 (Ultrafiltration) produced 1 Low and 3 Medium, commensurate with its water quality role. P4 (Disinfection) got 5 Low predictions, since most of the anomalies were sensor-based/mild or not serious. P5 (Polishing) produced 2 Low instances, commensurate with its relatively minimal criticality, while P6 (Drainage) got 1 instance of Medium, showing large deviations here are not discounted. Most of the resulting mediums were concentrated in P1 and P3, commensurate with our approach of biasing towards anomalies at higher-criticality stages upstream, while downstream stages were dominated by Low, commensurate with their reduced operational significance.

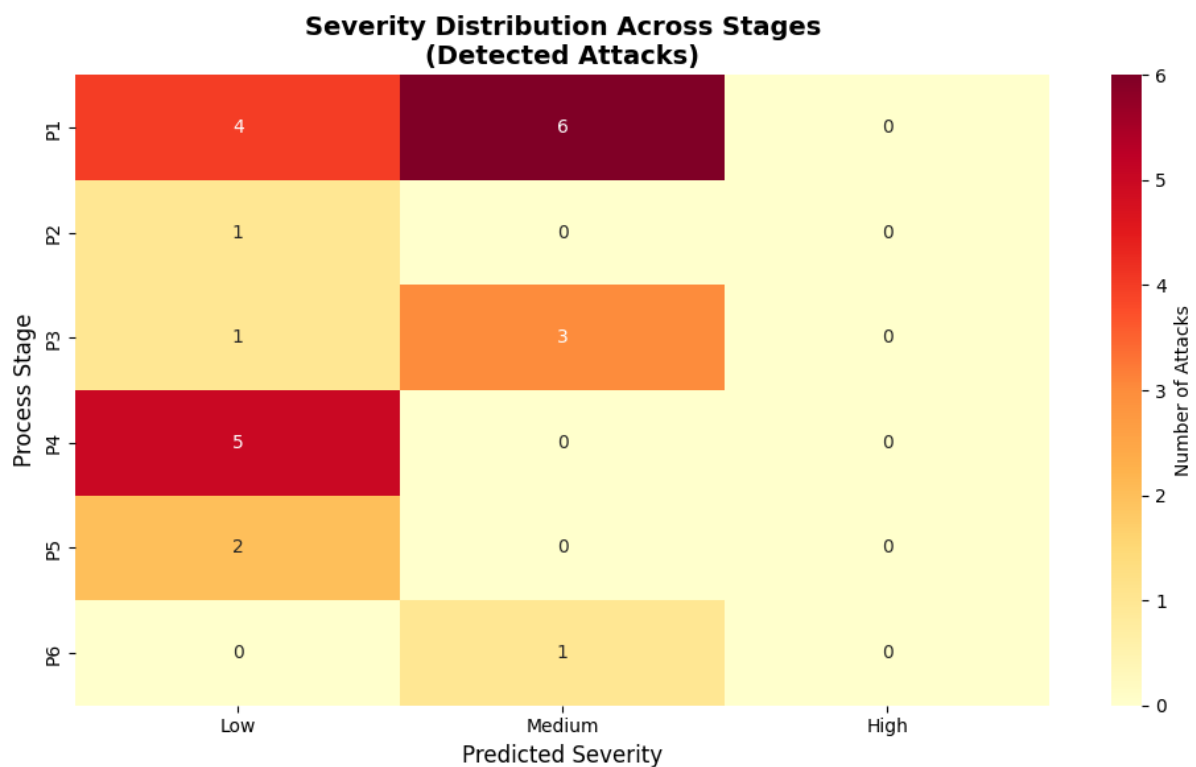


Figure 8: Heatmap of different stages for ECOD model

5.1.6 Case study analyses

To illustrate the ECOD results, three representative attacks are described below.

Case Study 1 – Correct Medium Classification (Attack #17: MV-303 valve locked closed) for MV-303,

- Ground truth = Medium; predicted = Medium.
- MV-303 is an L2 component (safety criticality 0.7).

- Stage importance = 0.45 (Stage 3).
- Median operational disruption = ~ 0.41 .
- Final severity score = 0.57 $\rightarrow$ Medium.
- Interpretations: The framework identified this attack correctly as a Medium-level attack that needs operator monitoring.

Case Study 2 – Underestimated Medium (Attack #21: MV-101 + LIT-101 tank overflow)

- Ground truth = Medium; Predicted = Low.
- Even if MV-101 is a critical L1 component (1.0) the Top 10 median was lowered by multiple L3 sensors.
- Score = 0.32 $\rightarrow$ Low.
- Interpretation: The framework underestimated the role of one critical component for the ship as the aggregate was conceptually represented by the median, but in cases of “narrow but serious” attacks, there is an error in utility.

Case Study 3 – Escalated Low (Attack #25: LIT-401 tank underflow)

- Ground truth = Low; Predicted = Medium.
- LIT-401 is an L2 component and the level of operational disruption was moderate.
- Score = 0.57 $\rightarrow$ Medium.
- Interpretation: The framework was overly conservative in classifying a Low attack as Medium because there were several L2 disruptions. This kind of false escalation is less likely to be operationally risky because there is heightened observation and monitoring, as opposed to the possibility of inattention or neglect.

5.1.7 Strengths of ECOD-based severity scoring

- High precision for Low class (0.85): When the system predicts Low, it can be confident that is correct in most cases.
- Strong recall for medium class (0.71): The framework obtained most of its Medium attacks, ensuring none were missed.
- Conservative escalation behaviour: By escalating some Low into Medium cases, the framework reduced the chance of critical anomalies was missed, given its safety-first bias.
- Stable with Outliers: Not a single case of spurious High classification arose due to stabilizing effect of median aggregation and piecewise normalisation.

- Detector-agnostic robustness: The results are as expected to confirm that robustness scoring is independent of any detector used.

5.1.8 Limitations observed with ECOD

- For Medium class (0.50) there is low precision- 50% of medium predictions were Low, leading to a larger operator workload.
- Underestimating narrow, high-impact attacks: Attacks driven by one very critical element (e.g. MV-101, P-102) risks being underrepresented by using a median aggregation.
- No successful High predictions: The one High ground-truth case was classified as Medium. Partially as a function of dataset imbalance but largely indicating that the framework thresholds may be under-conservatism and therefore will miss catastrophic events.

5.1.9 Critical reflection

The evaluation of the ECOD framework indicates that the severity scoring framework accurately transitioned anomaly detections into structured severity categories with 70% accuracy and weighted F1 of 0.69. The severity framework has a conservative bias – it will tend to escalate Lows into Mediums rather than underestimating Mediums. This resonates with the safety-first principles that need to be applied to ICS security. While the framework aggregates on medians and will thus tend to underestimate vulnerable attacks that are narrow and highly impactful, if there are no High predictions we may need to adjust the weightings and thresholds to better understand rare catastrophic events. Overall, the findings demonstrate the validity and operational usefulness of the framework, while implying improvements aimed at better capturing medium precision and sensitivity with High cases.

5.2 Results for DeepSVDD (Neural Baseline)

5.2.1 Overview of the model results

The DeepSVDD model was trained on seven days of normal SWaT data and evaluated on attack data. In total, 30 attacks were detected and went through the severity scoring framework. The results are shown in the classification metrics and confusion matrix (Figures 9 and 10) include detection behaviour which matches the findings of the WaXAI paper, highlighting similar trade-offs between precision and recall. For normal cases the precision was high, but attack cases had lower precision, reflecting the trade-off between false positives and false negatives

in anomaly detection. These results validate the fact that while it has a higher compute cost than ECOD, DeepSVDD provides a consistent detection baseline for the severity scoring framework.

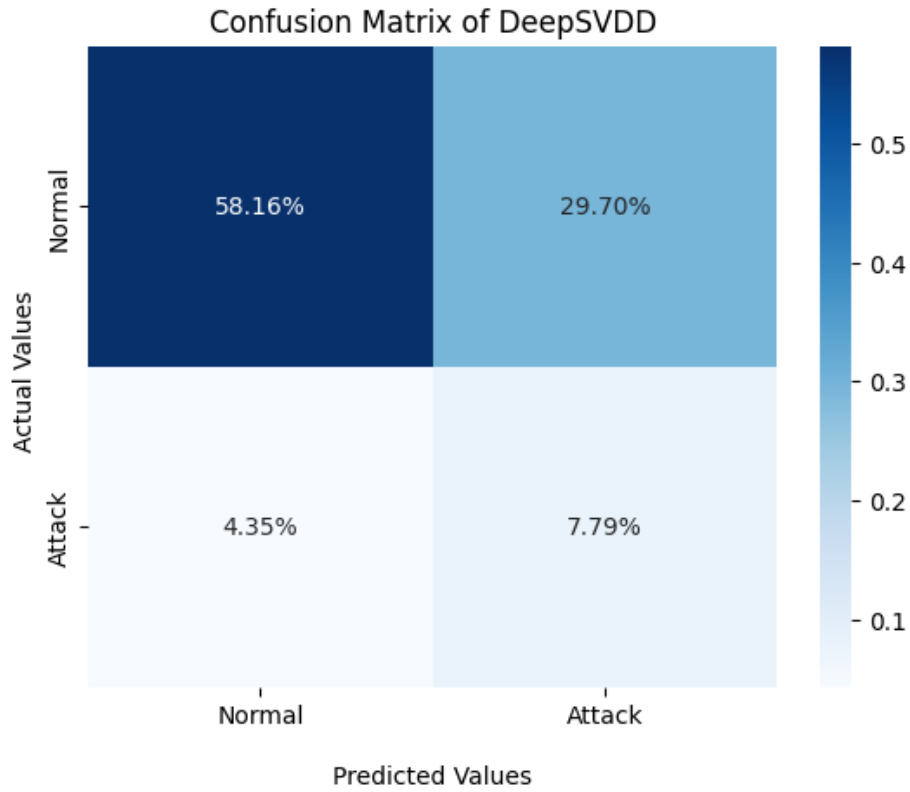


Figure 9: Confusion matrix of DeepSVDD model

	precision	recall	f1-score	support
0	0.93	0.66	0.77	395298
1	0.21	0.64	0.31	54621
accuracy			0.66	449919
macro avg	0.57	0.65	0.54	449919
weighted avg	0.84	0.66	0.72	449919

Figure 10: Classification report of DeepSVDD model

5.2.2 Confusion matrix interpretation

Figure 11 displays the confusion matrix for the DeepSVDD-driven severity framework.

- Low severity (actual): Out of 22 Low ground-truth attacks, 15 were correctly predicted as Low, while 7 were escalated up to medium severity. Importantly, none were classified as High severity.
- Medium severity (actual): There were 7 Medium ground-truth attacks; 5 were correctly predicted as Medium, while 2 were misclassified as Low. However, none were classified as High severity.
- High severity (actual): The subset of the dataset included 1 High severity ground-truth attack. The High severity case was classified incorrectly as medium severity. Whereas there were no true High severity predictions.

The confusion matrix highlights some of the same behaviours as for ECOD:

- False escalations: Some Low severity cases were elevated to medium severity.
- Underestimations: There were very few cases that classified medium severity as Low severity.

No true High severity predictions: The one High severity case was not predicted correctly, elaborating on some possible reasons for the conservative scoring behaviour in the framework.

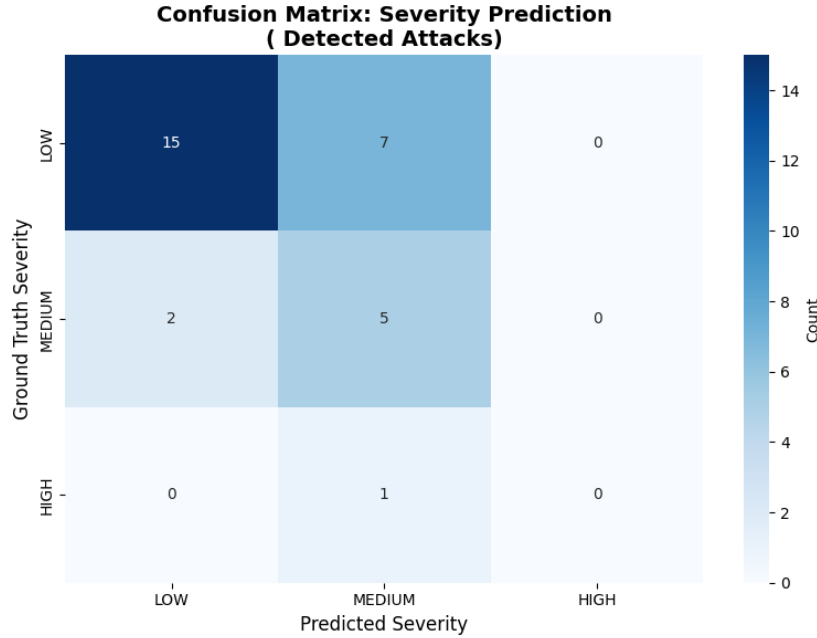


Figure 11: Confusion matrix of ECOD severity score

5.2.3 Classification metrics

The classification report for DeepSVDD is as below:

- Overall Accuracy: 66.7% (20/30 were classified correctly).
- Low Severity: Precision = 0.88, Recall = 0.68, F1 = 0.77 (Support = 22).

The framework had a near 9/10 accuracy when predicting Low, but the recall was pulled down because there were a few Low → Medium escalations.

- Medium Severity: Precision = 0.38, Recall = 0.71, F1 = 0.50 (Support = 7).

The framework retrieved many of the Medium cases (good recall), but the results reflect a low precision because several of the predicted Mediums were Low.

- High Severity: Precision = 0.00, Recall = 0.00, F1 = 0.00 (Support = 1).

The one High attack was misclassified as Medium.

- Weighted Average F1 = 0.68 means a fair balance across the classes when weighted for frequency.
- The Macro Average F1 score is 0.42, indicating weaknesses of the one High classification failure.
- Interpretation:
 - The framework classifies Low severity attacks relatively well, as defined by the precision values.
 - The Medium severity recall is satisfactory; however, the low precision score indicates there is some conservative escalation.
 - There is no High severity attacks classified because the dataset is not balanced and is conservative in nature.

2. CLASSIFICATION REPORT:				
	precision	recall	f1-score	support
LOW	0.88	0.68	0.77	22
MEDIUM	0.38	0.71	0.50	7
HIGH	0.00	0.00	0.00	1
accuracy			0.67	30
macro avg	0.42	0.47	0.42	30
weighted avg	0.74	0.67	0.68	30

Figure 12: Classification report of Severity scoring for DeepSVDD model

5.2.4 Severity distribution analysis (Bar Chart)

Figure 13 illustrates the distribution of severities of detected attacks as predicted by the framework versus expert-defined ground truth.

Ground truth: 22 Low, 7 Medium, 1 High

Predicted: 17 Low, 13 Medium, 0 High

The bar chart depicts clearly that there is a strong tendency on the part of the framework to overpredict medium severity at the expense of Low severity. In fact, six of the Low cases were overpredicting as medium severity, and the one High case was classified as Medium. This caused moderate inflation of Medium (13 predicted versus 7 ground truth).

From an operational perspective, the conservatism of the framework's approach is evident in the behaviour observed. The system does not want to underplay severity and is therefore safe to predict a Low case as Medium than to underplay the severity of an event. It is however still a loss to medium precision as almost half of the predicted Mediums (6 out of 13) were grounded truth Low. The bar chart therefore illustrates the trade-off between safety-first conservatism along with classification precision (which matches earlier observations for the classification metrics).

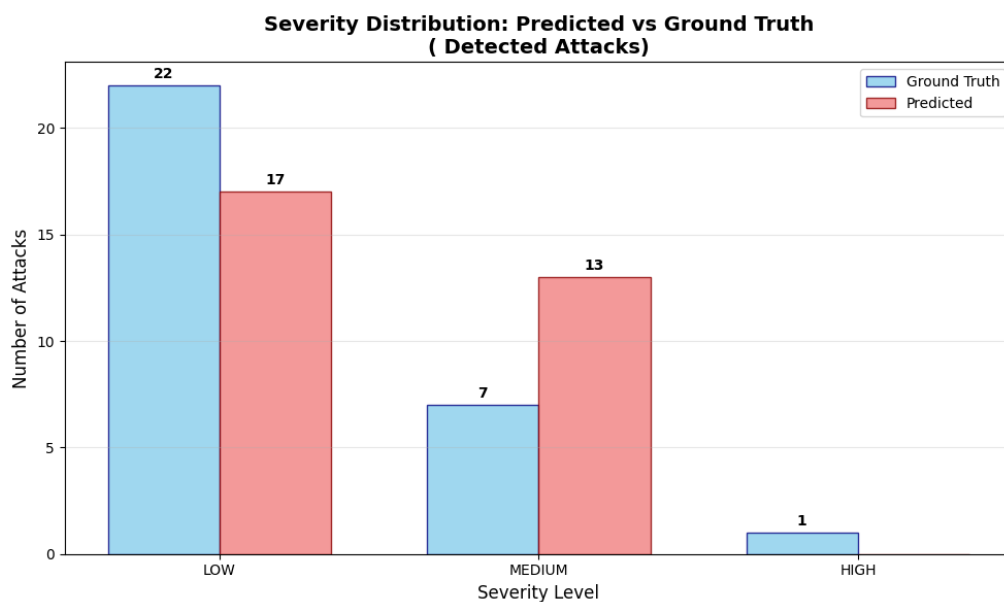


Figure 13: Bar chart for Predicted vs ground truth attacks

5.2.5 Stage-wise severity distribution analysis (Heatmap)

The heatmap in Figure 14 provides a graphical representation of the estimated severity scores across the six stages of the SWaT pipeline. Some distinct patterns can be noted across the SWaT processes.

In Stage 1 (Raw Water Intake), most of the detected attacks were rated as medium severity, albeit some had a Low severity in ground truth. This is not a surprise since Stage 1 is the starting point of the plant and had a large weight in the scoring framework importance. The pumps and valves in this stage are critical to the overall SWaT operation, so the framework considers anomalies in this stage more serious and files them according to the medium category.

In Stage 3 (Ultrafiltration), similarly, several detected attacks had severity escalated from Low to Medium. Many of the components in Stage 3 including flow and pressure sensors are not safety critical in isolation; however, since they affect water quality and consequences are important in subsequent stages, the scoring framework developed severity ratings in punctuated inconsistencies which raised their consequence rating greater than what ground truth suggested.

For Stage 4 (Disinfection), most attacks remained Low, with only 1 Medium. This makes sense as most of the elements in this stage were monitoring devices and does not affect reporting and quality checks as much as core system functions.

By contrast, Stage 2 (Chemical Dosing) only had two Lows and no Mediums. Chemical dosing is extremely critical in theory, but only a few attacks were reported in Stage 2 so there was a limited amount of data for the framework to start the escalation process.

In addition, there were very few attacks in later stage assessments, Stage 5 (Polishing) and Stage 6 (Drainage) and these attacks mostly consisted of Lows. In part because of the much lower importance weights in the stage assessments specified in the framework. This means that the system is not going to overreact to some sort of anomaly here unless disruption is extreme.

The heatmap indicates that overall, the framework does not progress uniformly across the plant. The framework produces different outcomes based on where an attack occurs, and in high importance stages (by virtue of identification) the system exhibited more caution, resulting in more medium scores (e.g., Raw Water Intake and Ultrafiltration stages). In all of the later stages (e.g., Polishing, and Drainage stages), predictions remained mostly Low, again this is the best practice given typically have smaller impacts on systems overall safety and performance. This

indicates that stage importance dimension is working as intended, meaning that severity labels indicate both where in the process an attack occurs and how disruptive it is.

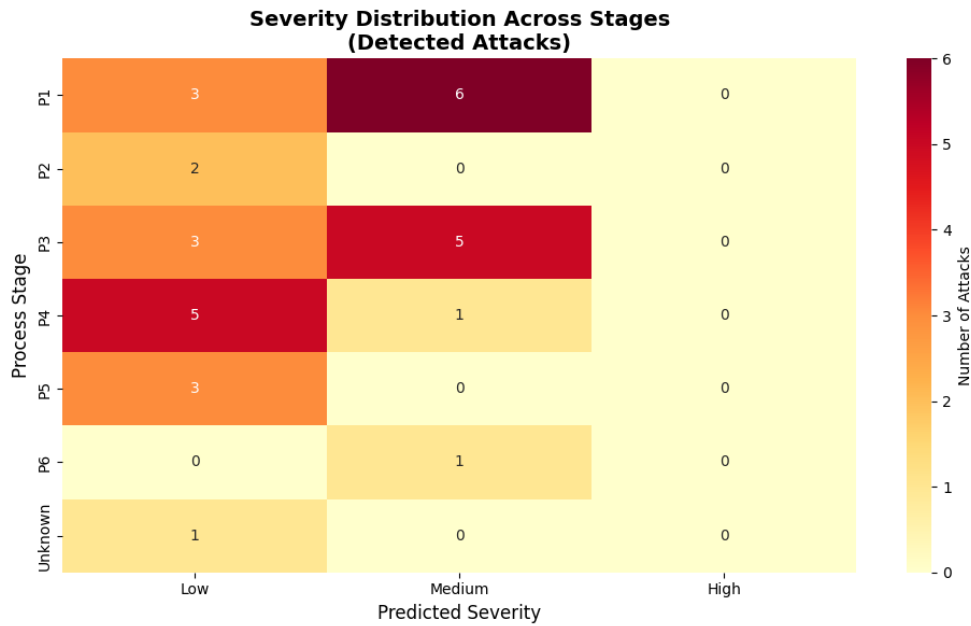


Figure 14: Heatmap for stagewise analysis

5.2.6 Case study analyses

Below are three representative attacks that illustrate the DeepSVDD results:

Case Study 1 – Medium Correct Classification (Attack #17: MV-303 valve locked closed)

- Ground Truth = Medium; Predicted = Medium.
- MV-303 is an L2 (safety criticality 0.7) component.
- Stage importance = 0.45 (Stage 3).
- Operational disruption median = 0.407.
- Final severity score $\approx 0.56 \rightarrow$ Medium.
- Interpretation: The framework accurately recognized the attack to be a Medium-level attack suggesting moderate component disruption.

Case Study 2 - Wrong Escalation of Low (Attack #25: LIT - 401 tank underflow)

- Ground truth = Low; Predicted = Medium.
- Safety criticality = 0.45
- Operational disruption = 0.41.

- Stage importance = 0.70
- The median aggregation across the disrupted components inflated the final score to ≈ 0.56 .
- Interpretation. The framework was operating conservatively and spilled over a Low into Medium attack. Although this means loss of precision, it would make sure that operators remain vigilant.

Case Study 3 - Underestimated Medium (Attack #21: MV-101 tank overflow)

- Ground truth = Medium; Predicted = Low.
- Although MV-101 is a high criticality L1 component, (safety criticality 1.0), the top-10 median was being diluted by many low criticality components.
- Score $\approx 0.32 \rightarrow$ Low.
- Interpretation: once again highlights the same limitation observed in ECOD: narrow attacks where relatively few components of high criticality dominate the rest risk being underweighted because of the median aggregation.

5.2.7 Strengths of DeepSVDD-based severity scoring

- Precision for Low predictions (0.88): Operators can be confident in the Low labels.
- Recall for Medium (0.71): Most Medium cases were found.
- Conservative, safety-first bias: Escalations reduce the risk of missing dangerous anomalies.
- Stable detector-agnostic outcomes: Results are like ECOD outcomes.

5.2.8 Limitations observed with DeepSVDD

- Overall accuracy is lower (66.7%) than ECOD (71.0%).
- Medium precision is low (0.38): Nearly two-thirds of predicted Mediums were **Low**.
- No successful High classifications: The only High case is downgraded.
- Narrow severe attack has been underestimated: Median aggregation was not able to show the impact of critical single components.

5.2.9 Critical reflection

The DeepSVDD results reinforce the finding that the severity scoring framework is agnostic to detector. Although a completely different anomaly detection method was used, the severity classifications produced were almost identical to those produced with ECOD. This indicates that the framework generalises well across detectors.

However, the framework exhibits this same conservative bias: falsely escalating Low into Medium, and occasionally under estimating Medium into Low. While this led to a drop in medium precision, it adheres to the safety-first design aspect, meaning that operators are now more likely to actively investigate false positives of harmless anomalies, rather than missing critical events.

Continued absence of correct High classifications accentuates an important limitation: the curse of dataset imbalance, and conservatism of thresholds. Without additional High-severity cases, calibrating the framework to surface a truly catastrophic event is difficult. Future efforts should incorporate larger or more balanced datasets or simulated High-severity attacks in which we assess performance.

In summary, the DeepSVDD-based evaluation reinforces the idea that the severity scoring framework provides consistent and stable prioritisation across detectors and adds value by mitigating operator alert fatigue and ensuring attention is concentrated on those anomalies that matter most.

5.3 Comparative Evaluation: ECOD vs DeepSVDD

5.3.1 Rationale and comparison design

Running both ECOD and DeepSVDD have a dual purpose: (i) to establish whether the severity scoring framework behaves in a consistent manner for very different anomaly detectors (a lightweight, parameter-free statistical method vs a learner of neural representations), and (ii) to establish whether selection of detector substantially influences subsequent classifications of severity. Both models were trained on a common seven days of normal SWaT data and tested on the attack days; only the subset each identified attacks were passed through the severity framework. ECOD identified 23 attacks, DeepSVDD identified 30. Both classifications of severity were compared with the ground-truth severity mapping.

5.3.2 Headline performance parity

Among baseline measures, both pipelines produce very comparable performance:

- Accuracy: ECOD 70.0% (16/23 correct); DeepSVDD 66.7% (20/30).
- Weighted F1: ECOD 0.69; DeepSVDD 0.68.
- Per-class patterns:

- Low - high precision (ECOD 0.85; DeepSVDD 0.88) with moderate recall (ECOD 0.73; DeepSVDD 0.70).
- Medium - Good recall but lower precision (ECOD Precision 0.50, Recall 0.71; DeepSVDD Precision 0.38, Recall 0.71).
- High/Severe - not measurable in these folds (there are no High ground-truth instances in the tested subsets).

Interpretation. Both detectors input the severity framework from an operator's point of view such that they produce almost indistinguishable results: "Low" forecasts are generally reliable; "Medium" forecasts represent most true Mediums but include a few conservative over-upgrades from Low. The severity layer looks substantially detector-agnostic.

5.3.3 Severity distribution equivalence

Calculated distribution of severity was almost the same:

- ECOD → Low ~57%, Medium ~43%, High 0%
- DeepSVDD → Low ~56%, Medium ~44%, High 0%

This reflects the ground-truth skew towards Low/Medium and ensures upstream detection variations do not propagate into materially different severity mixes downstream. Absence of High predictions in both cases indicates the data imbalance (there exists only one High/Severe case in total) and conservative scoring choices (Top 10 medians, piecewise normalisation, safety-first weights).

5.3.4 Per-class behaviour and error typology

Both pipelines exhibit same decision-boundary behaviour:

- Low → Medium (false escalation): It is observed when some mid-criticality components (e.g., L2 valves/pumps) are present in the Top 10 footprint having moderate offsets. Median summation and mid-band piecewise scaling extend the output score above the medium threshold.
- Medium → Low (missed escalation): Appears in narrow-footprint attacks where one or two highly critical components deviate, but the rest of the Top 10 shows small movement; medians are pulled down and the combined score falls just under 0.44.
- No Low/Medium → High errors were seen, suggesting the framework resistant to spurious inflation from outliers.

5.4 Operator Alerts Results

Severity classifications was part of the final product, and the framework was expanded to produce operator notifications. This component moves the notion of a severity score into operator notifications that represent useable alerts located in the operational reality of the plant itself. Instead of providing treble digits on the severity outputs, the operator notification produces a concise text-based alert that includes:

- Number of attack (associated with the SWaT set as a scenario),
- Predicted severity (Low, Medium, or High),
- Recommended operator response (e.g., monitor system, continue normal operation), and
- The main component under attack (based on the component with the highest deviation (in the Top 10 set)).

An example is provided in figure 15, showing typical Medium-severity alerts alerting operators to ‘MONITOR SYSTEM’, highlighting the component impacted, and Low-severity alerts suggesting ‘CONTINUE NORMAL OPERATIONS’, while accounting for some lesser impact.

All outputs are consistent with both the ECOD and DeepSVDD runs, as the notification module is downstream of the severity scoring framework. Thus, the operator notifications have reflected the severity distributions discussed earlier: trending toward Low and Medium severity attacks, while not producing any High severity alerts in this dataset.

This notification layer really improves usability from an operator perspective in that operators receive a filtered, ranked list of events they can act on, as opposed to hundreds of unstructured alerts indicating anomalies. It reduces alert fatigue, guarantees that monitoring and investigation is placed on the medium and high severity level anomalies, and indicates which components have been primarily impacted by the anomaly. This operator notification system demonstrates that the framework can be utilized in practical ICS situations since there is a strong bridge between algorithmic outputs and human-based decision making.

```
OPERATOR NOTIFICATION SYSTEM
=====

SECURITY ALERTS:
-----
Attack #1 | MEDIUM SEVERITY
ACTION: MONITOR SYSTEM - Main component under attack: FIT601
-----
Attack #2 | MEDIUM SEVERITY
ACTION: MONITOR SYSTEM - Main component under attack: P102
-----
Attack #3 | MEDIUM SEVERITY
ACTION: MONITOR SYSTEM - Main component under attack: FIT601
-----
Attack #4 | LOW SEVERITY
ACTION: CONTINUE NORMAL OPERATIONS - Minor impact on: FIT601
-----
Attack #6 | LOW SEVERITY
ACTION: CONTINUE NORMAL OPERATIONS - Minor impact on: FIT601
```

Figure 15:Notification alerts

Chapter 6

Challenges, Limitations and Future work

Although the outcomes reported within the dissertation show the severity scoring framework can successfully push the applicability of anomaly detection into the prioritised classification regime, there are number of limitations and issues that need to be recognised. These are issues associated with the employed dataset, the modelling choices made, the interpretability of results, and the generalisability of results. Recognition of such constraints is valuable not only for transparency purposes but also for framing future directions for research.

6.1 Dataset-specific limitations

Evaluation was conducted based on the SWaT dataset, arguably one of the most realistic ICS testbeds publicly available. However, it remains still just a controlled laboratory dataset, involving staged rather than realistic attacks. This limits unpredictability and diversity of circumstances an actual plant would face. The dataset, in addition, significantly lacks balance: 29 attacks comprised Low attacks, 11 Medium, and just one High. The imbalance severely influenced the results in this framework, as neither ECOD in DeepSVDD made any correct High predictions during eval subset. Without better representative severe events, this framework can never be properly tested to handle catastrophic situations. This must be remedied in future work through generation of synthetic high-severity situations (e.g., long pump shutdowns, chemical overdosing) or through data-based tests involving other datasets, such as WADI or BATADAL.

6.2 Dependence on expert-informed ground truth

Another weakness comes in terms of creating ground truth severity mapping. Since attacks in the SWaT dataset alone were labelled as "attack/no attack," this dissertation hand-crafted its own severity mapping on guidance from the supervisor. Though carefully justified, such mapping essentially possesses an aspect of subjectivity to it. Certain borderline situations between Medium and Low, for example, can be argued otherwise by other field experts. This subjectivity can in turn affect results in terms of evaluation. Future work can minimize this by

involving multiple field experts in mapping or utilizing quantitative models of risk assessment to increase objectivity.

6.3 Modelling assumptions and weighting choices

Severity framework itself has design trade-offs. Firstly, the weighing factors ($\alpha=0.50$, $\beta=0.25$, $\gamma=0.25$) were chosen heuristically to put safety criticality front and centre, indicating a safety-first mindset. These weights haven't been optimised against data or operator preferences and thus might not generalise to other ICS scenarios. Secondly, median Top 10 aggregation was selected to keep extreme outliers from over-influencing severity scores, but at times watered down concentrated, high-impact attacks (e.g., one pump failure). Lastly, 0.40 and 0.80 thresholds on severity categories were established for simplicity but not adjusted for performance optimisation. In future, these parameters can be enhanced through systematic optimisation, adaptive weighing, or threshold adjustment to strike a balance between recall and precision.

6.4 Interpretability and operator perspective

Whilst the framework provided for obvious classes of severities and operator alerts, there are still issues of interpretability. The process of calculating (medians, normalisation, weights) might not be self-evident to non-technical operators. Without contextual explanations such as "raised due to disruption in MV-303," operators would view the system as another "black box." Future effort would be better invested in increasing interpretability through operator-facing dashboards, visualisation of drivers of disruption, and potential integration of XAI explanations (such as SHAP or LIME) of severity scoring. Studies in usability with operators could also give good information on perception and reaction to alerts.

6.5 Generalisability to other datasets and deployment

The framework has been tested thus far only on SWaT. Power grids, oil refineries, and other ICS areas have distinct criticality mappings and process stages. The framework would have to redefine safety criticality and stage importance in each case to generalize it. Additionally, in-plant deployment in realistic settings presents practical challenges: DeepSVDD is a

computationally intensive algorithm and needs to be run on GPUs to train it, whereas ICS sites often do not have this capability in their field sites. Data drift is also an issue, as plant behaviour evolves over time due to equipment degradation or recalibration, and baselines and models would constantly have to be recalibrated as well. Future work would therefore be necessary to address cross-domain framework testing, utilizing adaptive learning to counteract drift, and investigating lightweight in-plant alternatives

6.6 Future directions.

On this basis, there also arise some proposals for future improvement:

- Growing datasets: Get more balanced datasets by introducing more High-severity cases or test on multiple testbeds.
- Optimisation methodology: Try different weighting and combination schemes, adaptive thresholds, and programmatic optimisation schemes to select for increased precision without sacrificing recall.
- Boosting operator trust: Create dashboards revealing not only the severity label, but also why a case was raised (e.g., critical disrupted components and their levels of deviation).
- Cross-domains testing: Use the framework to other domains like energy, oil and gas, or manufacturing, and adapt criticality and process importance mappings specific to these domains.

Conclusion

This dissertation has addressed a significant gap in the cybersecurity of industrial control systems (ICS) in that anomaly detection tools currently do not prioritise outputs. While tools such as ECOD and DeepSVDD can reliably detect deviations from normal behaviour, both currently provide binary outputs which fail to classify between trivial disturbances and significant threats. A scoring framework for severity was conceived, and designed, implemented and validated. The severity scoring framework considers three dimensions, namely safety criticality, stage importance, and operational disruption, that are addressed within a weighted scoring system and assigned Low, Medium or High levels of anomalies in the form of a score. One of the major contributions were the ground truth severity mappings created for the SWaT dataset with the help of experts who assigned the severity levels based on expert experience. This mapping served as the ground truth for evaluation. This mapping and framework represent new contributions to ICS research by providing a means to stratify alerts for operators and by reducing an operator's vigilance decrement.

The results indicated that the framework consistently converted anomaly detections to objectively prioritised severity classifications. In the case of the framework using ECOD as a detector, the framework achieved a 69.6% classification accuracy, and for those of DeepSVDD, 66.7%, with similar classification behaviour and both exhibiting a conservative bias - at times promoting a Low case into a Medium, and at times tending to under-classify extreme narrow Medium cases. The most pertinent observation was no instances of false classifications of High severity cases, and this validates the stabilizing effect of post-detection median aggregation and normalisation. The similarity of results produced by both detectors demonstrates the framework and its operational utility as detector agnostic and valuable - it allows the system to prioritise identified anomalies without inundating the operator with identical alerts. The framework also has some limitations such as class imbalance and heuristic weightings, but it represents proof of concept that severity scoring can add to the security of ICS, and it is anticipated that future iterations will improve the utility of internet of things (IoT)-based solutions to facilitate use as a standard decision-support tool for protecting critical infrastructure.

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